

CNS

Computer

Network

Security !!

• Computer Network :

- A comp. n/w is a grp of interconnected computers.
- It allows comp. to communicate with each other & to share resource & info.
- The best CN is the Internet.

• Data Communication.

- It means exchange of data betⁿ two devices.

• Componentⁿ of data commⁿ.

- (i) Sender : sends data/info.
- (ii) Receiver : receives from sender
- (iii) Transmission medium : physical path from sender to receiver.
- (iv) Protocol : programmer & n/w are connected together by certain rule
- (v) Message : It is actual data & info to be sent.

• Goals of Comp. N/w:

- Resource sharing
- High Reliability
- Inter-process commⁿ
- Flexible access of shared resources
- Security
- Performance & Scalability

• Types of n/w:

① PAN.

- Personal Area N/w (Private)
- It is for to a single person.
- It often n/w range of 10 to 100 m. from person to device.

② LAN

(Private)

- Local Area N/w
- It is most frequently used n/w
- Range is 2km.
- eg. home, school, office, building

③ Man

(Private or public)

- Metropolitan Area N/w
- Much larger than LAN but smaller than WAN
- eg. town, cities. (5-50km)

④ WAN

(Private or public)

- Wide Area N/w
- eg. country, cities. (100,000 km)

* OSI Reference Model.

- The ISO was one of the first organization to formally define a common way to connect computers. Their order, called the Open System Interconnection (OSI)

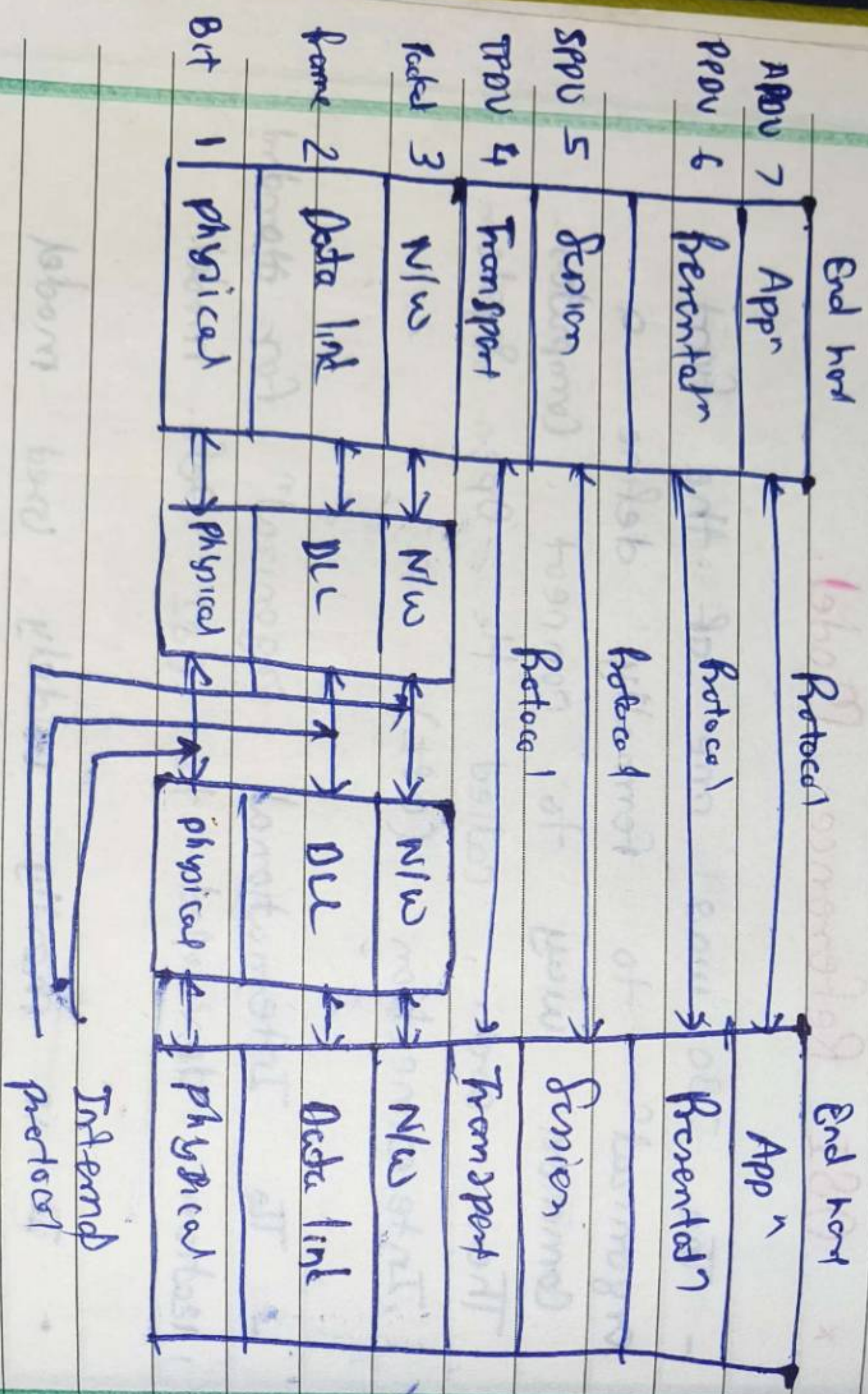
- The International organization for standardization developed the OSI ref. model.
- It is mostly widely used model for networking

Date : _____

- It is seven layer std.

• OSI provides following services :

- ① Peer-to-peer logical services.
- ② std ~~com~~ for commⁿ betⁿ system.
- ③ Defines pt of interconnection for the exchange of info. betⁿ system.



OSI Model

Date : _____

• Functions of OSI layers :

① Physical layer :

- It transmit a bit stream over a communication channel.
- It deals with procedures & fn required for transmission.

• Pⁿ :

① It considers the characteristics of interface betⁿ devices & transmission media.

② Representation of bits :

- Physical layer encodes the bit stream into electrical or optical signals.

③ Data rate : Duration of bits

④ Synchronization of bit :

- The transmission rate & receiving rate must be the same.
- This is done by synchronizing clocks at sender & receiver.

③ Data link layer.

- It is used for transmitting frames from one node to the next.

• fn :

① Framing : The frames received from the network layer are divided into manageable data units called frames.

② Flow Control : Physical addressing

③ Error Control

④ Access Control : multiple devices are connected to same link.

④ Network layer

- It is used for delivery of packets from the source to destination.

• fn :

① Logical addressing

② Routing : Network layer route or switch the packets to its final destination in an internetwork.

④ Transport layer

- It is used for delivery of message from one process to another.

• fn :

① Port addressing

② Segmentation : message divided into parts

③ Connection Control

Transport layer performs connection-oriented services with the destination machine.

④ Flow Control

⑤ Error Control

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⑤ Session layer :

- It is a m/w dialog controller.
- It establishes & synchronizes the interaction betⁿ commⁿ system.

• Pⁿ

① Dialog Control :

- Commⁿ betⁿ two processes take place in either half duplex or full-duplex mode.
- It layer manages dialog ctrl for this commⁿ.

② Synchronizatⁿ

- It adds synchro. pt^s into stream of data.

⑥ Presentation layer.

- It deals with syntax & semantics of the info. being exchanged.

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• Pⁿ :

① Translation.

- It maintains interoperability betⁿ two encoding systems.

② Encryption

③ Compression

⑦ Application layer.

- It is responsible for accessing the p/w by user.
- It provides user interface & other supporting services such as email, other file transfer, sharing db, etc.

• Pⁿ.

① File Transfer, Access & Management (FTAM)

② Mail Services

③ Directory Services.

• Connection Oriented Services

- Direct path is established between source & destination.

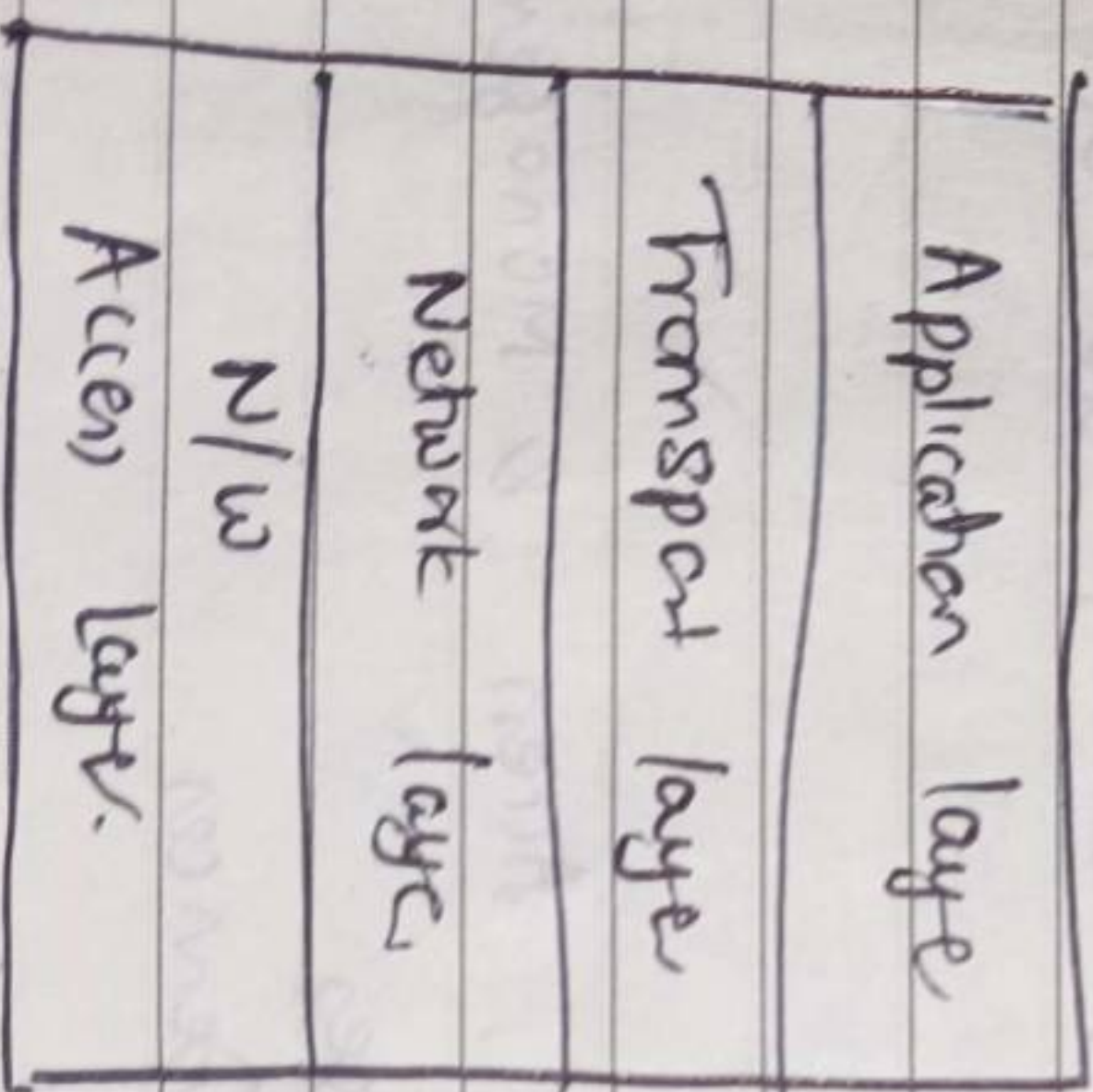
• Connectionless Services

- It goes directly from an idle condition into a data transfer mode

* TCP / IP Protocol

- It stands for "Transmission Control Protocol / Internet Protocol"

- It is a set of protocols that allow commⁿ across multiple diverse n/w.



- N/w Access Layer
- It is a grp of appliedⁿ requesting n/w commⁿ.

- It is responsible for generating the data & requesting requesting connections.

- N/w or Internet layer
- This layer parallel the IP & OSI's n/w layer.

- It defines protocol which are responsible for the logical transmission of data over entire n/w

- The n/w layer protocol (IP, ARP, ICMP) handle m/c to m/c (machine) commⁿ.

• Transport layer :

- This layer is responsible for data - blishing the connection b/w sender & receiver & also performs the task of dividing the data from the appⁿ layer into packets

- TCP : Transmission Control Protocol
 - It establishes connected "bid" source & destination (connected oriented)
 - UDP : User Datagram Protocol
 - It is responsible for identifying errors, & other tools during the transmission of info.
- Application layer
 - It includes all process & services that use transport layer to deliver data.

App'n Protocol : FTP, SMTP, SNMP

TCP/IP

OSI

- | | |
|-------------------------------------|---------------------------|
| ① 7 layers | ① 4 layers |
| ② mostly used | ② low in usage |
| ③ Vertically approached | ③ horizontally approached |
| ④ Delivery of package is guaranteed | ④ It is not guaranteed |
| ⑤ less reliable | ⑤ more reliable |

OSI

TCP/IP

- | | |
|-------------------------------------|---|
| - It has 7 layers | - It has 4 layers |
| - low in usage | - mostly used |
| - Vertically approached | - horizontally app. |
| - less reliable | - reliable |
| - Delivery of package is guaranteed | - Delivery of package is not guaranteed |

• Addressing

- TCP/IP Protocols uses 4 levels of addresses

① Physical Address

- also known as MAC (Media Access Control) or link address.

- It is defined by LAN or WAN
- Used by data link layer
- It is a unique address
- Size 48 bits (6 bytes)

② Logical Addresses

- also referred as IP (Internet Protocol) address, is an universal addressing system

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- Used in n/w layer.
- Type of IP addr.

① IPv4 - 32 bits

② IPv6 - 128 bits

③ Port addresses.

- It is used to assigned label to Process in TCP/IP.

- 16-bit

④ Specific Addresses

- It is designed by user for some appⁿ

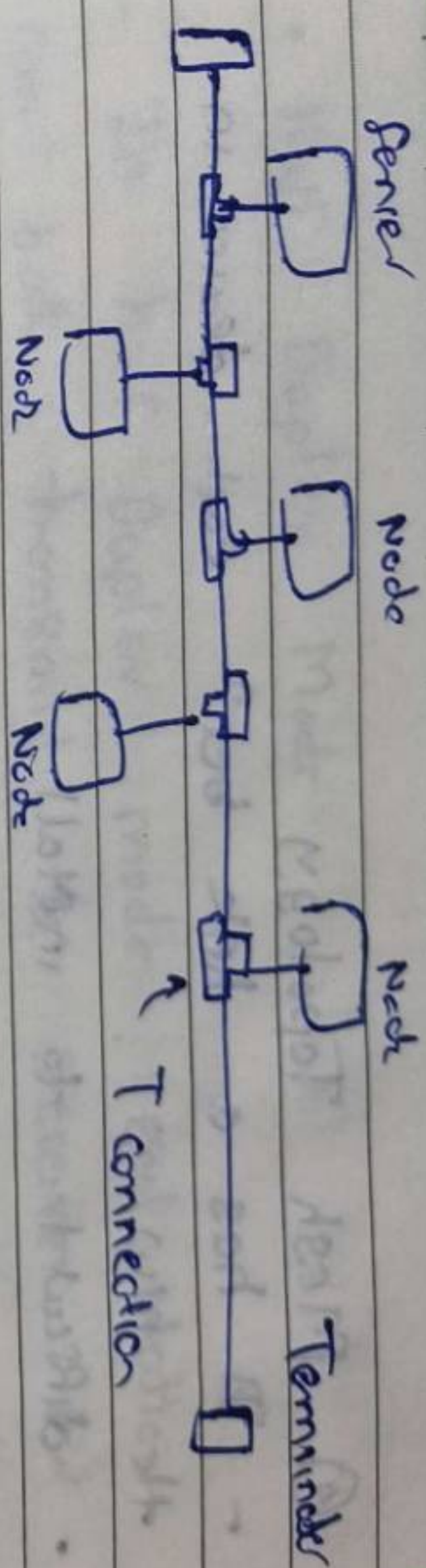
- eg. e-mail addr.

• Network Topology

- It is arrangement of diffⁿ elements like nodes, links & devices in a CN.

① Bus Topology

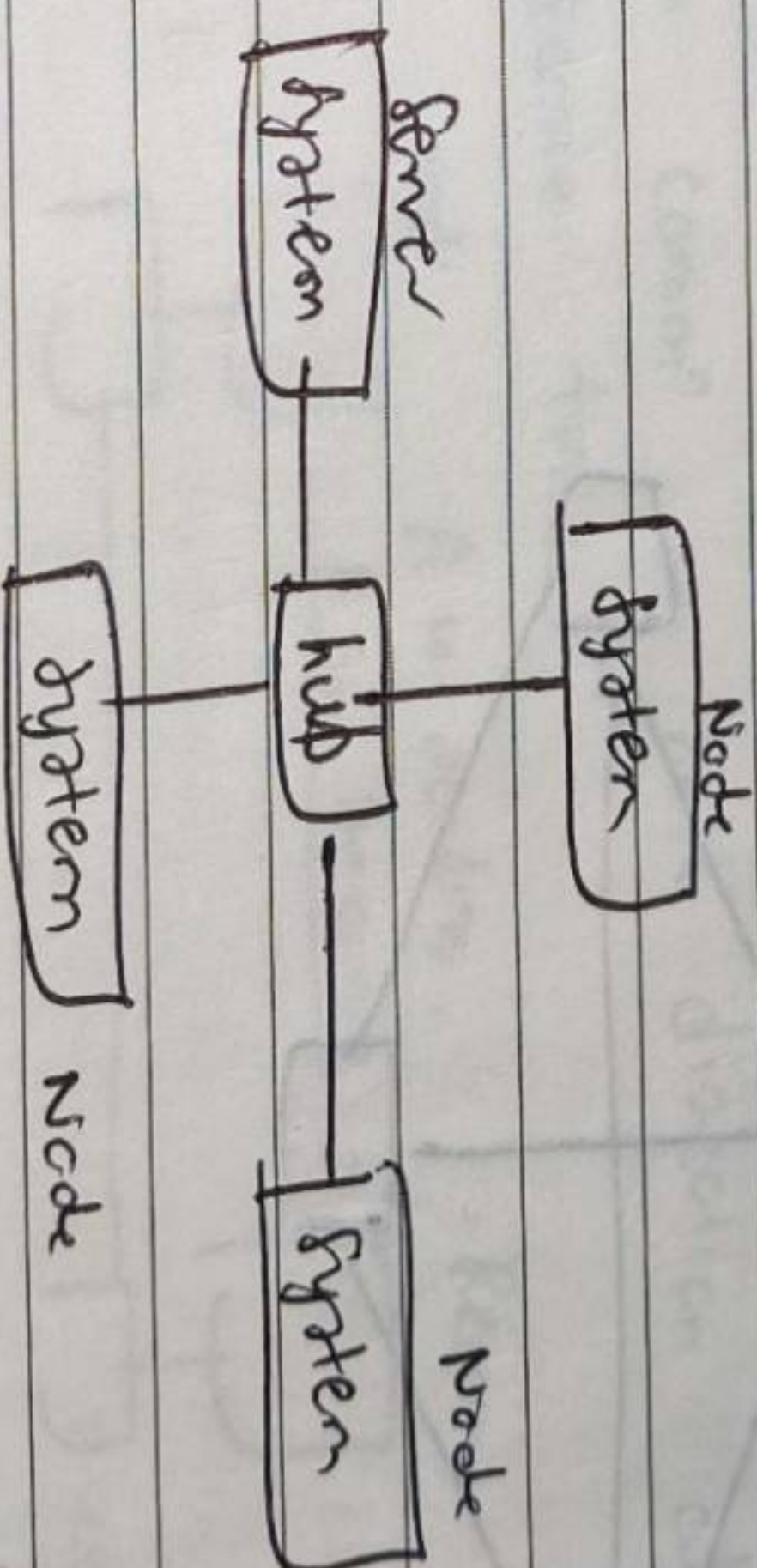
- also called horizontal topology
- multiple devices are connected only by one cable



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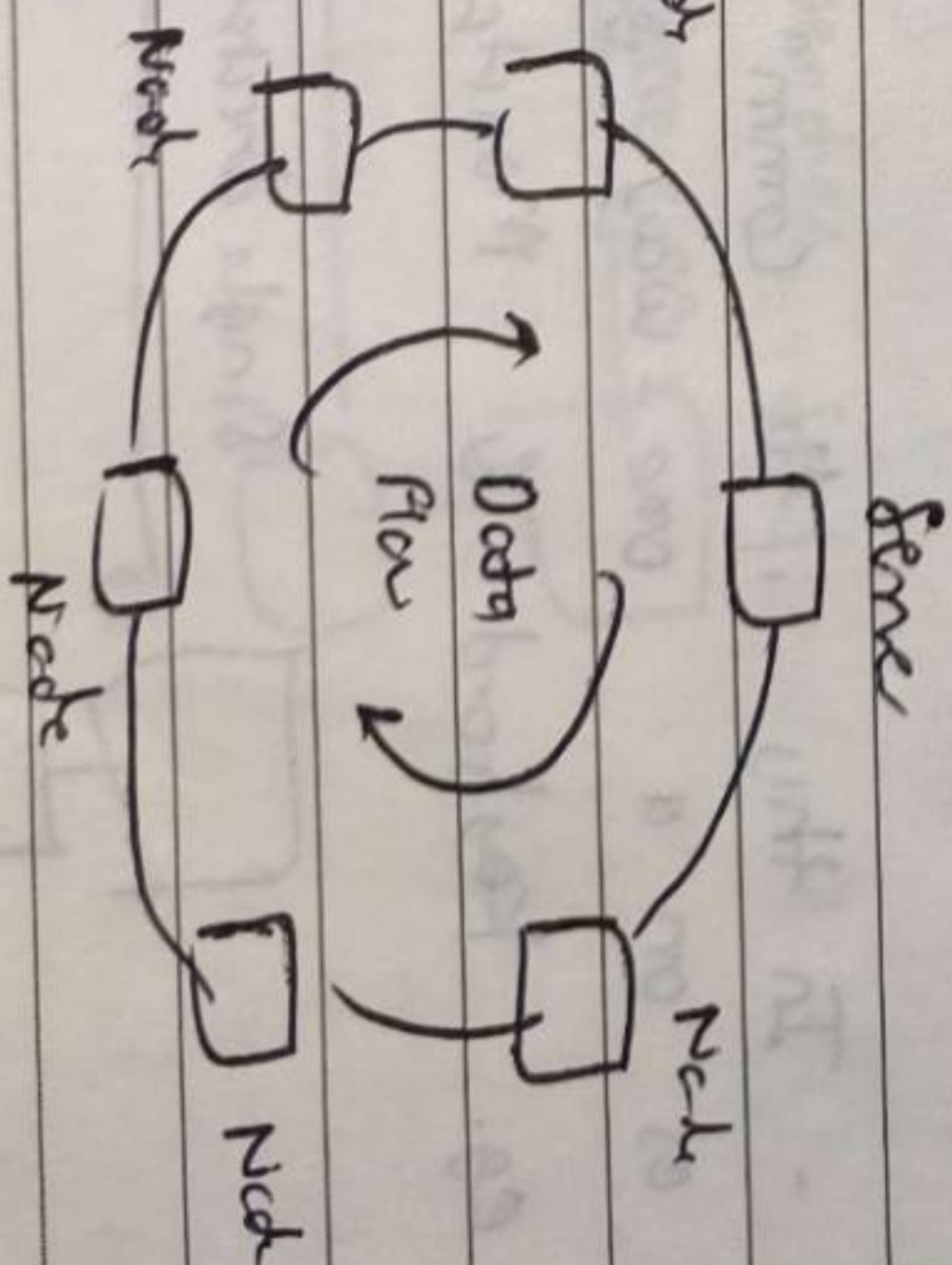
② Star Topology

- Multiple devices are connected to a single hub through a cable



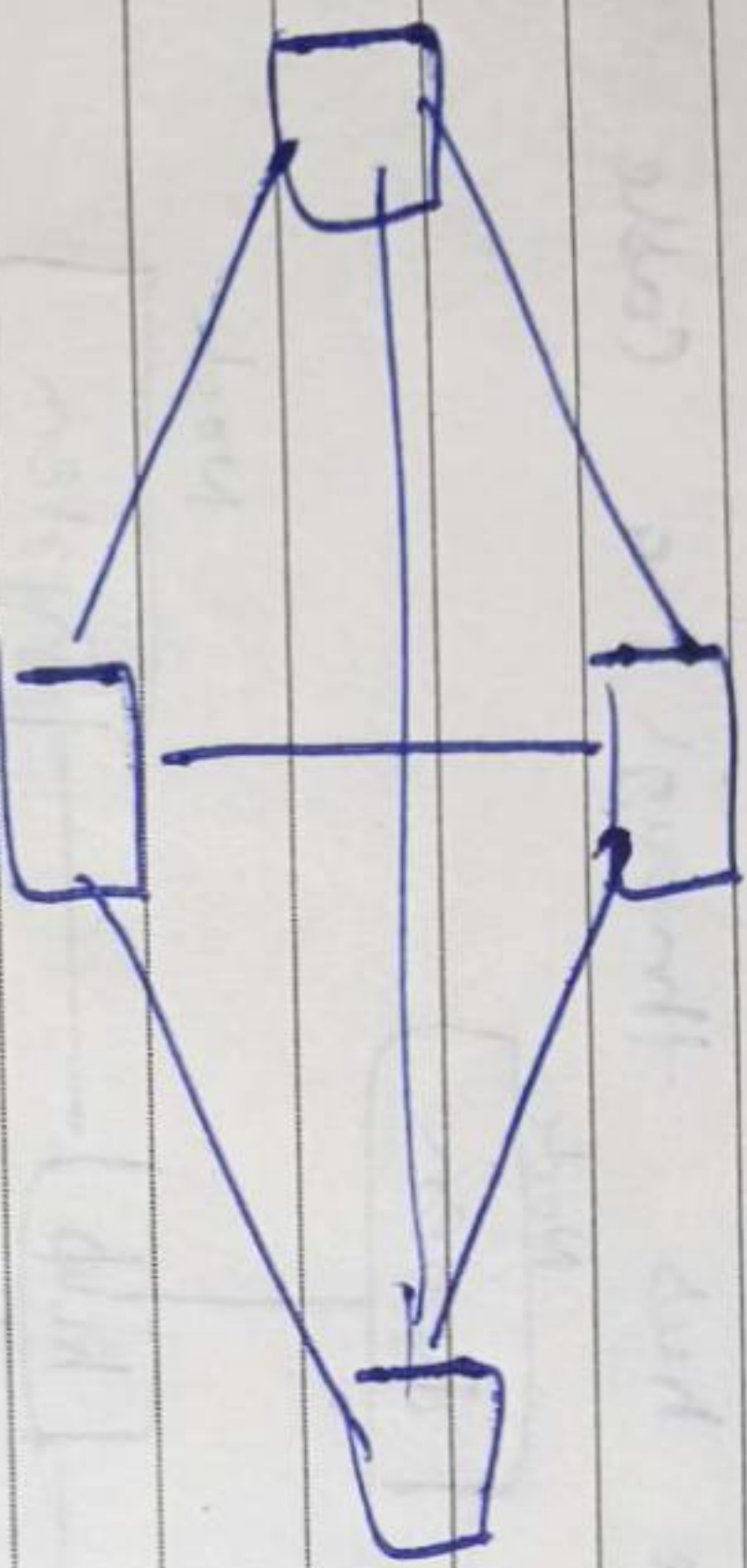
③ Ring Topology

- Each comp connected to next & last connected to first.



④ Mesh Topology

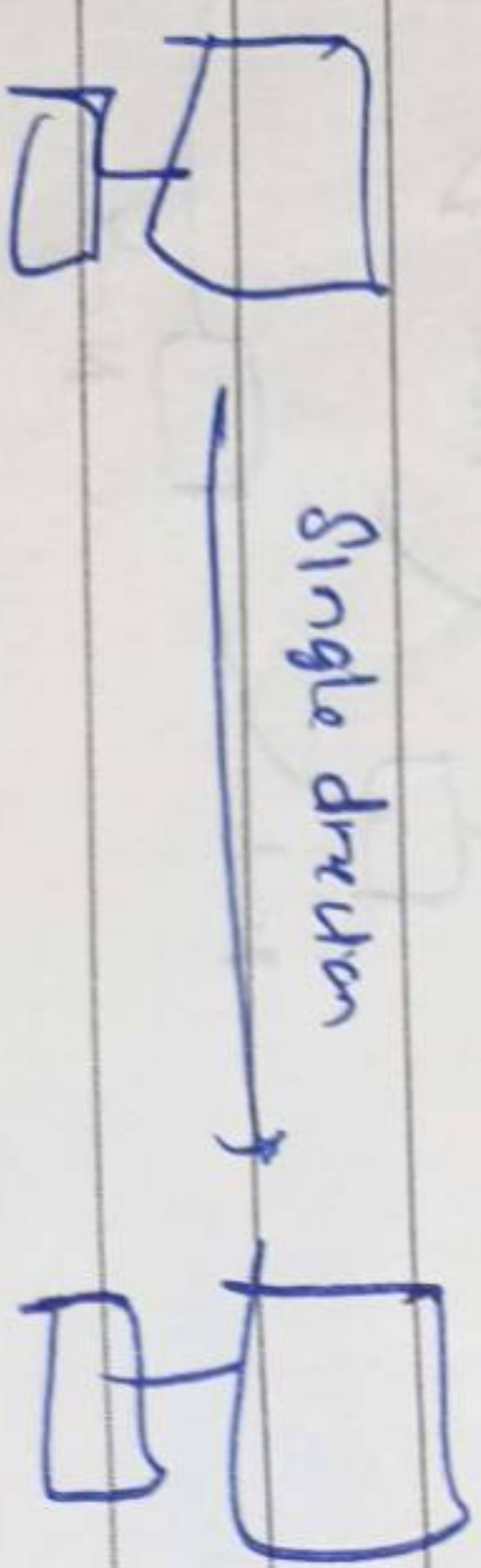
- It has a link betⁿ each device in the n/w.
- difficult to install
- easy to troubleshoot
- Every device connected with each other



• Simplex Mode

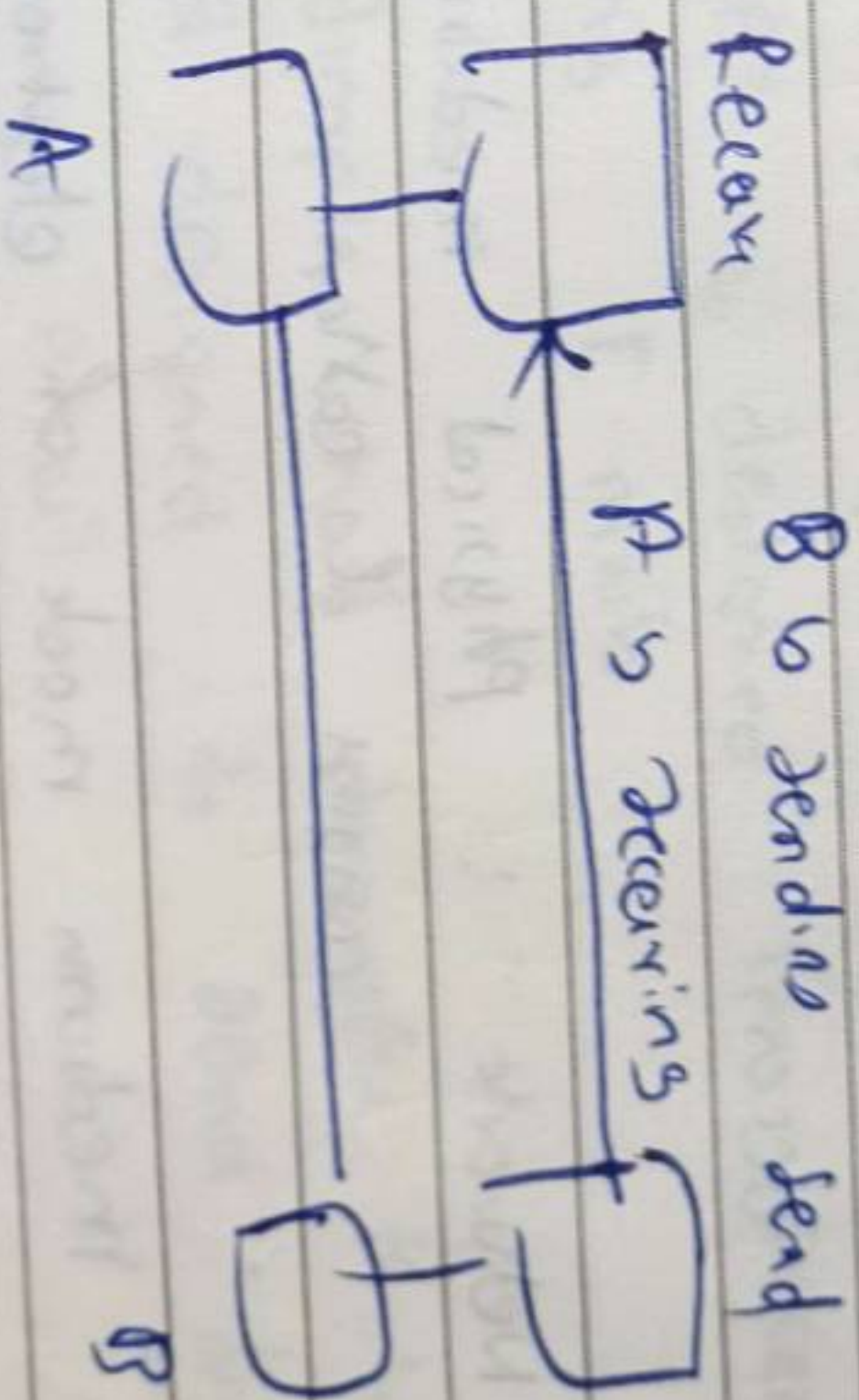
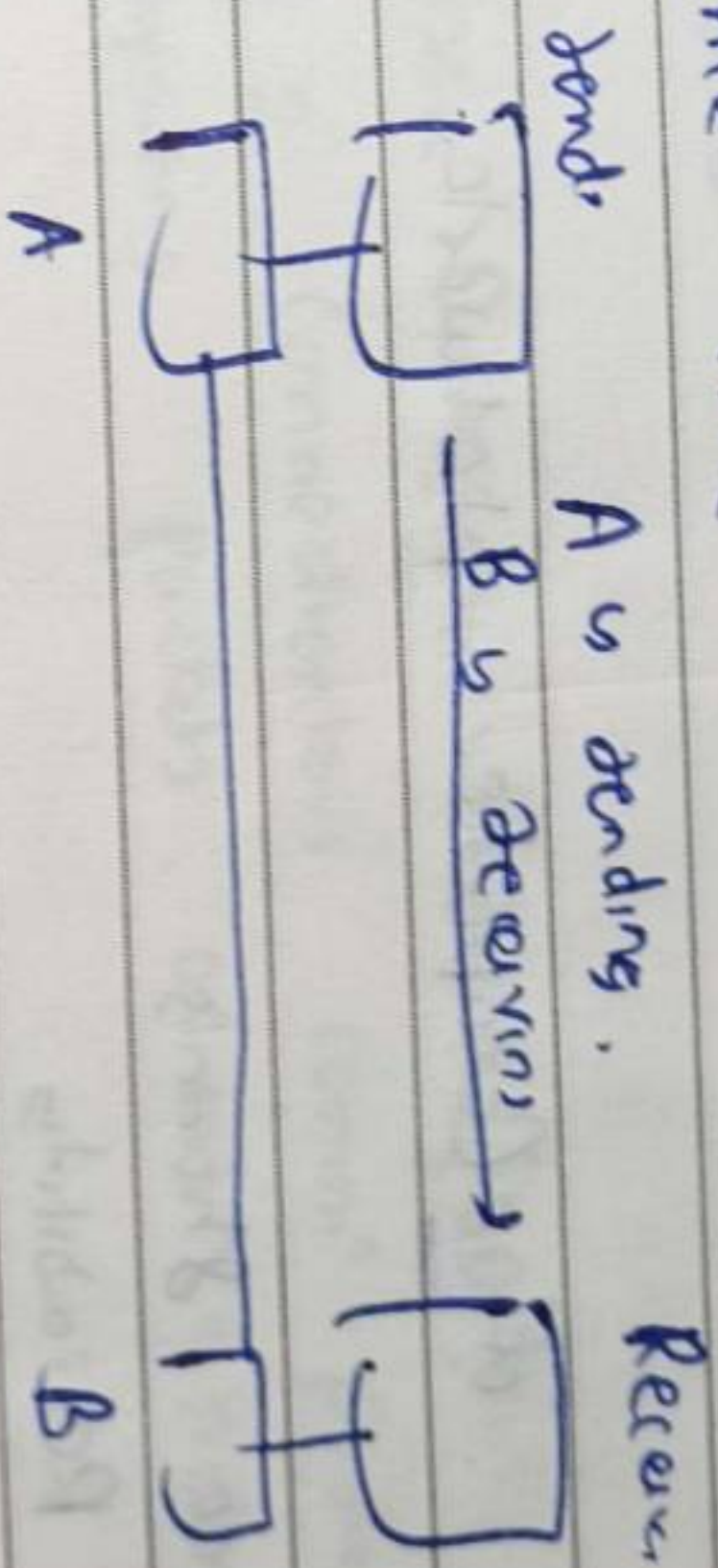
- In this, the commⁿ is unidirectional.
- eg. on a one-way street.

eg. keyboard & Monitor



• Half Duplex Mode

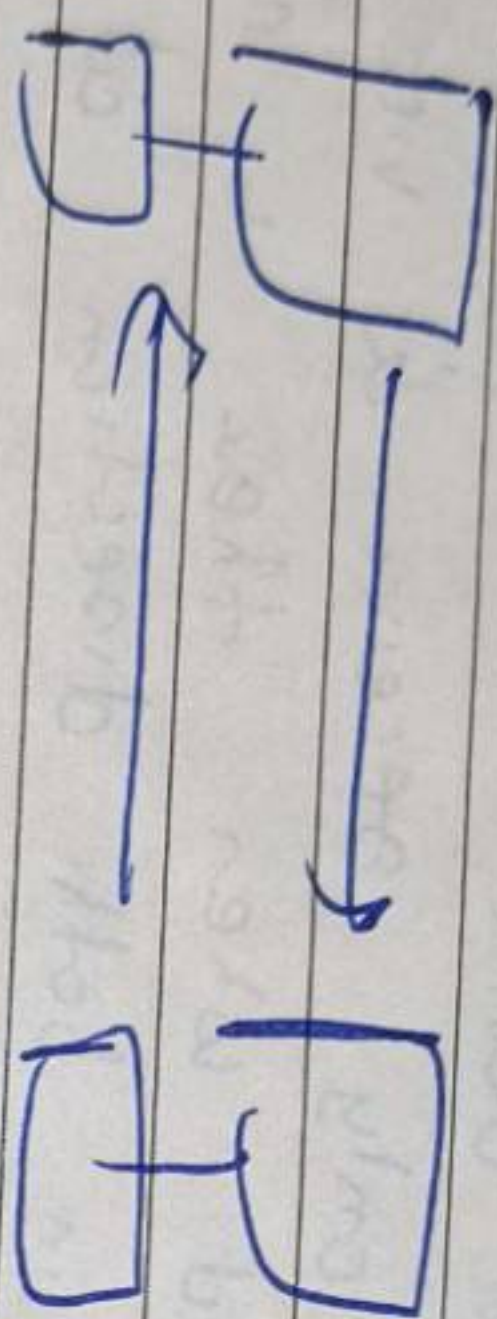
- In half duplex mode, each station can both transmit & receive, but not at same time.
- When one device is sending, the other can only receive & vice versa.
- It is used when there is no need for commⁿ in both direction at the same time



eg. Walkie - Talkie.

• Full Duplex Mode :

- In Full duplex mode, both stations, can transmit & receives simultaneously, eg. Telephone n/w.



• Goals of Computer Networks

- (i) Resource sharing
- (ii) High Reliability
- (iii) Inter-process Comm.
- (iv) Flexible access,

• Wired Network : Physical medium consist of cables

- Wireless : medium made of electromagnetic waves & Infrared waves (antenna & receiver)

• N/w Protocol

- (i) Transmission Control Protocol (TCP)
 - It is used for communications over a n/w

- (ii) Internet Protocol (IP) : It is the method or protocol by which data is sent from one comp. to another on internet.

- (iii) User Datagram Protocol (UDP)
 - It is connectionless comm. protocol for transporting packets across network.

- (iv) Post office Protocol (POP) :
 - POP3 is designed for receiving incoming E-mails

- (v) Simple mail transfer Protocol (SMTP)
 - It is designed to send & distribute outgoing email

(vi) File Transfer Protocol (FTP)

- transfer files from one to other machine
- e.g. Program files, multimedia, text files & docs.

(vii) HTTP (Hypertext Transfer Protocol)

- It is a set of common rules for client-server communication.

(viii) HTTPS (HTTP Secure)

- more secure extension of an HTTP
- In HTTPS, the browser & server establish a secure, encrypted connection before transferring data.

• Analog signal vs Digital signal

Q. It is used to communicate. (i) It transmits data info. in a continuous manner. (ii) It transmits data info. in a discrete manner.

(i) Continuous signals (ii) Discrete signals

(iii) Continuous signals (iv) Discrete signals

e.g. Human voice, e.g. computer, natural sound, optical drives & other electronic devices.

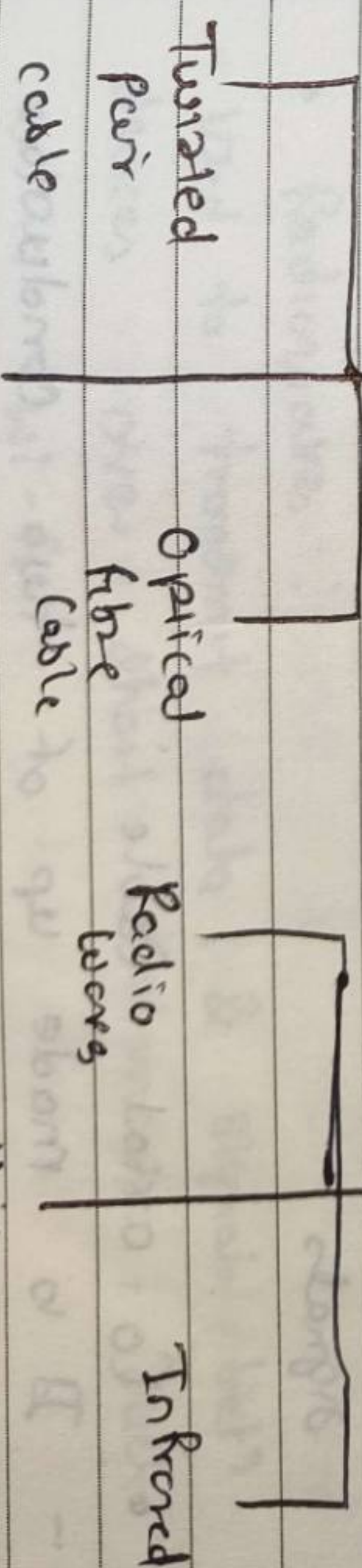
• Transmission Media

Types

Guided Media

Media

Media



Cable Micro waves

• Twisted Pair Cable (TP)

- It is a type of n/w cable that consists of pairs of copper wires that are twisted together.
- A wire pair acts as single communication link.
- Twisted pair types

① Unshielded TP cable

- UTP is a set of twisted pair of cable within a plastic sheath.
- It is ordinary telephone wire
- less expensive.

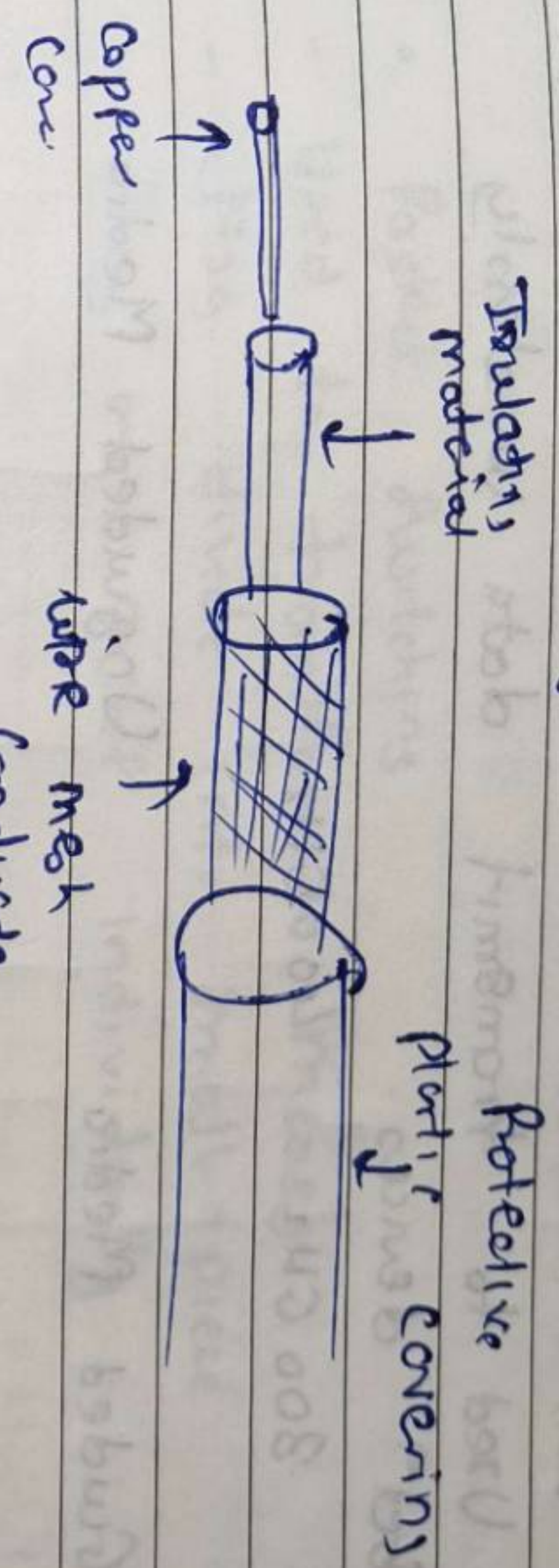
② Shielded TP cable

- It offers protective shielding around copper wires
- TP used for both analog & digital signals

• Co-axial Cable :

- It is made up of two conductors that share the common skin.

- It consists of a hollow outer cylindrical conductor that surrounds a single inner wire conductor.
- It is used to transmit both analog & digital signals.



- high bandwidth

• Fibre optic cable

- A Fibre optic cable is a light pipe which is used to carry a light beam from one place to another

• Radionwaves :

- Used to transmit data & signals between devices over short distances, such as in Wi-Fi
- 2.4 GHz & 5 GHz.

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• Microwaves.

- Used for long-distance data transmission & wireless n/w
- range 30GHz - 300.0 GHz.

• Infrared

- Used to transmit data wirelessly betⁿ devices.

- 300 GHz - 400 GHz.

Guided Media

Unguided Media

Used for point-to-point communication
Used for broadcast communication

• Switching Techniques.

① Circuit Switching :

- It used to establish dedicated path betⁿ sender & receiver.
- end-to-end path connection.
- e.g. telephone n/w & voice transmission.

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• Message Switching

- In this msg is transferred as complete unit.
- there is no establishment of dedicated path betⁿ sender & receiver

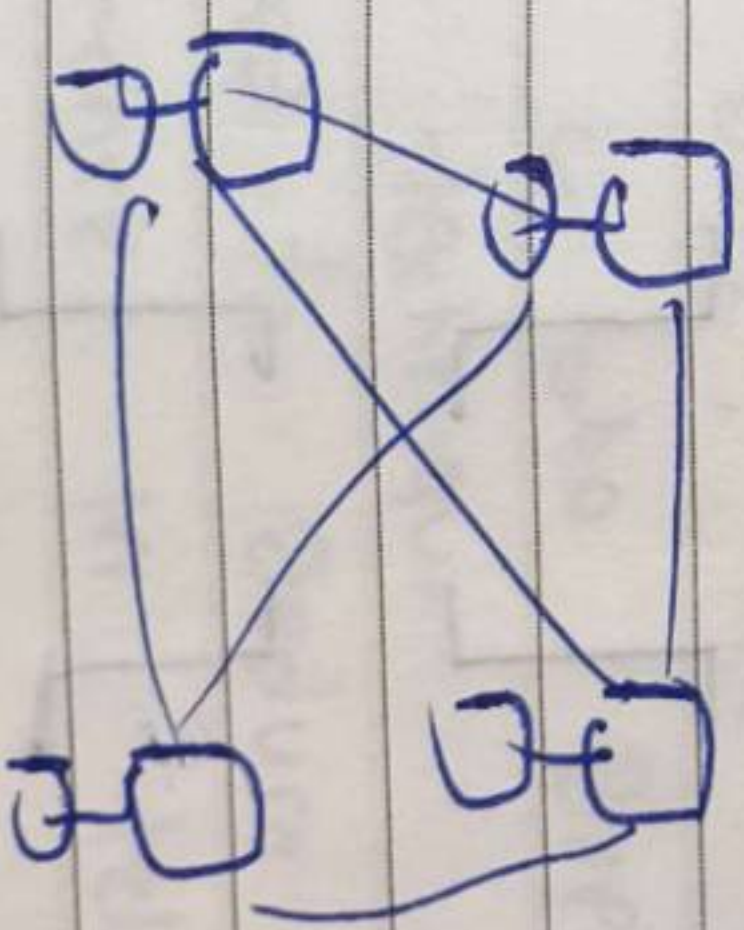
• Packet Switching

- Used to forward msg.
- Msg divide into small piece
- they are sent individually

• Network Architecture

① Peer-to-Peer n/w

- In this, all comp. are linked together with equal privilege.
- No dedicated server.
- Used for small environment.



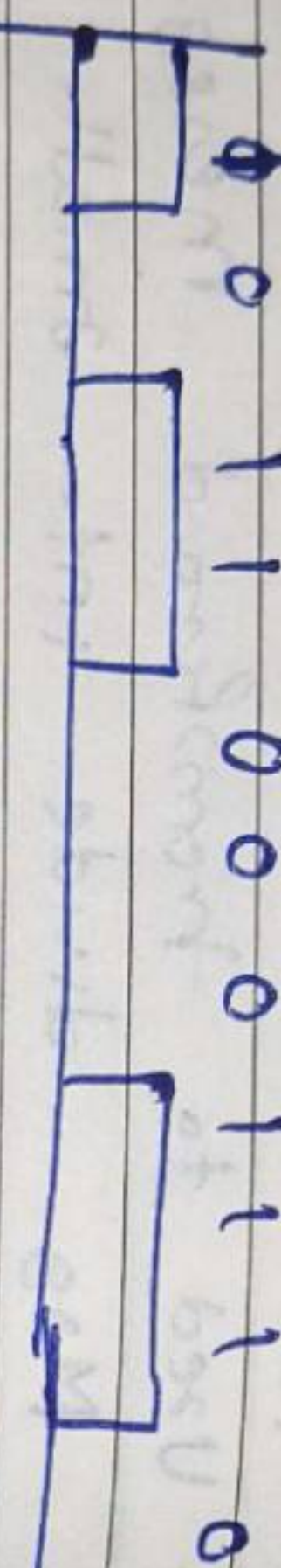
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① Client / Server N/w.
- Transfer data to client to server

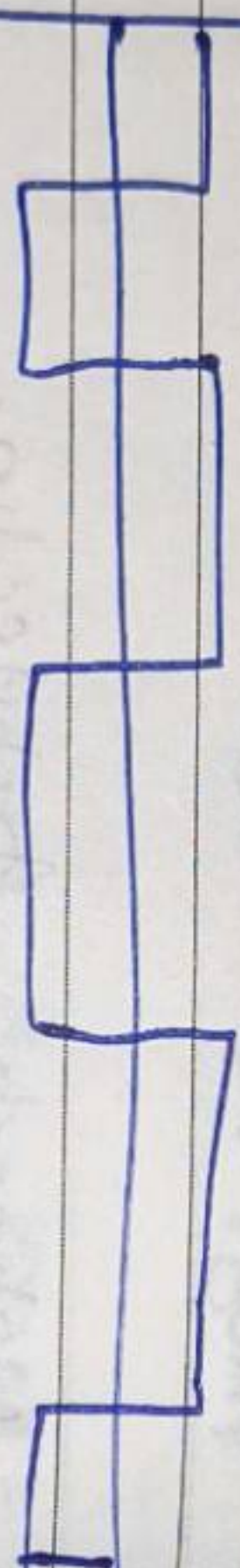
- Line coding

- It is process of converting bit stream into digital signal

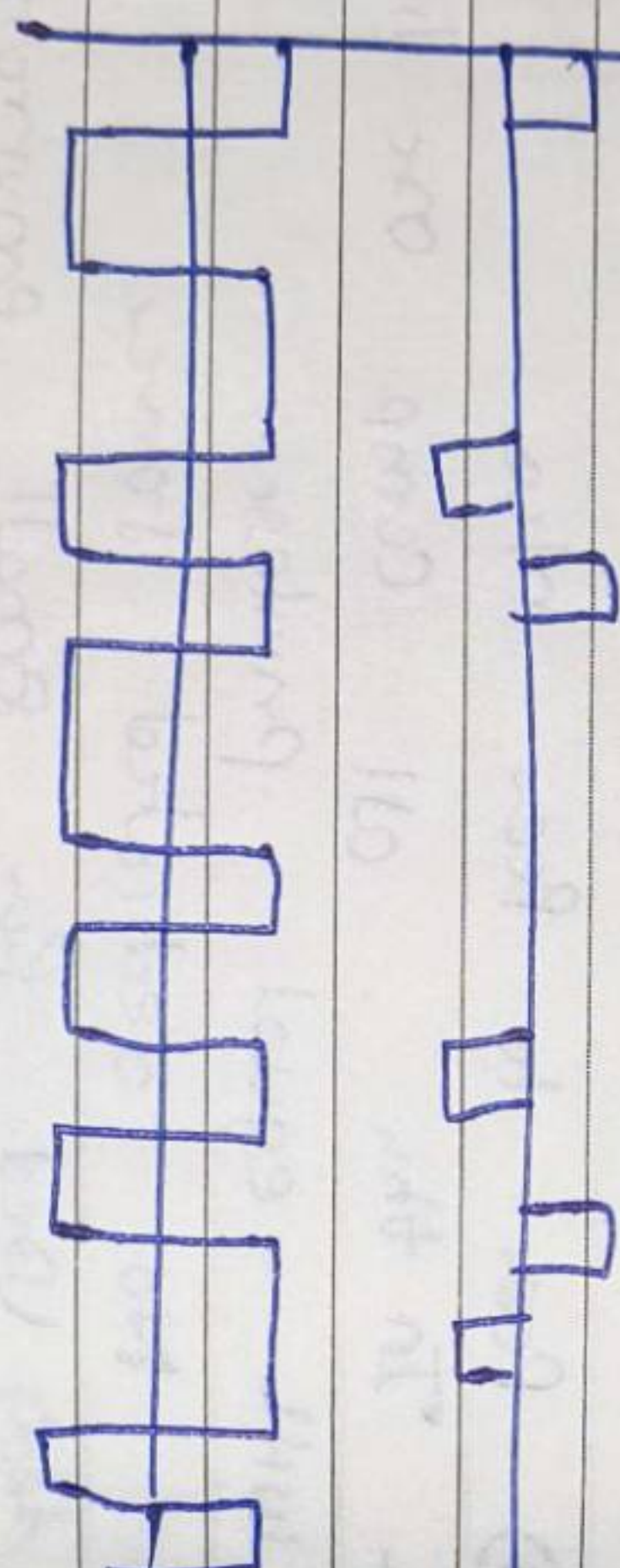
Unipolar RZ



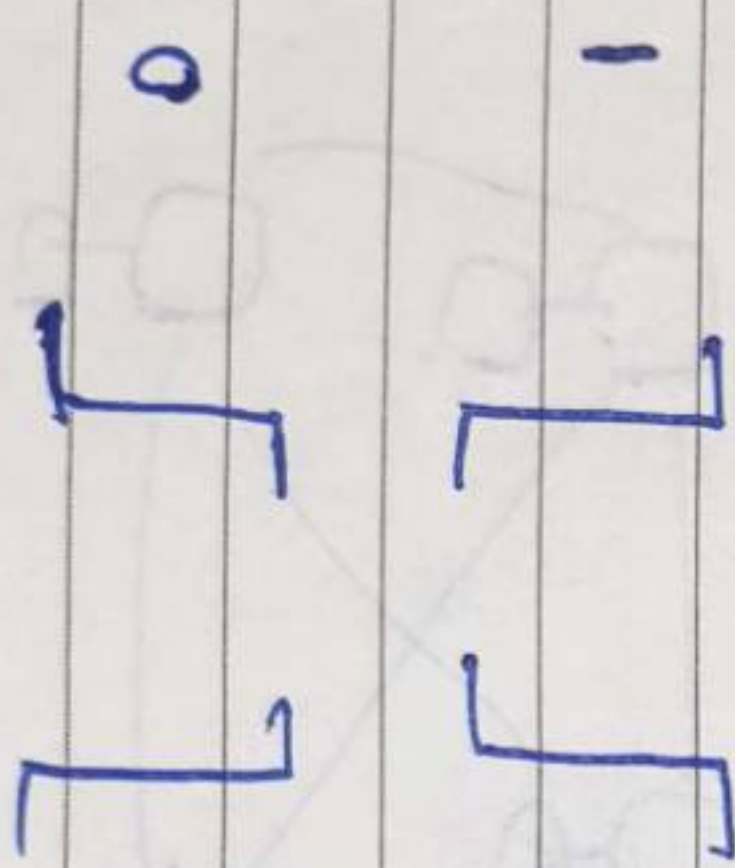
Polar NRZ



Manchester



Manchester



Date : _____

- Spread Spectrum

- It is a method of transmitting radio signals over a wide range of frequencies.

- FHSS

- Freq. Hopping Spread Spectrum

- It allows us to utilize bandwidth properly & maximum.

- In this tech, the whole available bandwidth is divided into many channels & spread betⁿ channels, changed continuously.

- DSSS

- Direct Sequence Spread Spectrum

- In DSSS, the bandwidth of original signal is also expanded by a diffⁿ technique.
- Used to reduce overall signal interference in telecommunication.

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• Network Device

- Also known as networking h/w, or physical devices that allow h/w on a comp. n/w to communicate & interact with each other.

• Hubs :

- All n/w require a central location to bring media segment together. These central location are called hubs.
- It is basically multiple repeated
- Hubs cannot filter data, so data packets sent to all connected devices.

- Passive hub : No power source is needed to connect devices without amplification.

- Active hub : This hub amplifies incoming signals before broadcasting & requires external power. Thus, it acts as repeater.

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- Intelligent Hub : It works like an active hub & includes some management capabilities.

• Repeater :

- It is an electronic device.
- It operates only in physical layer.
- Purpose is to extend the distance of LAN.
- Single port repeater
- Multi port repeater
- Smart repeater
- Optical repeater

• Bridges :

- It operates in both physical & data link layers.
- It extends the max. distance of n/w by connecting separate n/w segment.
- It performs data link fn such as error detection, frame formation & frame routing.

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• Switch

- It is a n/w comp. that connects devices on n/w & allows them to share info.
- Data link layer.

• Router

- It is a n/w device that connects devices & n/w to the internet, & to other n/w.
- It consist of combⁿ of n/w & s/w.

• Gateway

- It connects two independent n/w
- It is Protocol Converter.
- It operates all 7 layer of OSI
- It connects two n/w or appⁿ by translating data from one protocol to another.

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• NIC - N/w Interface Card

- Called as "N/w adapter".
- It allows computer to send & receive data.

• Ethernet

- It allows devices to communicate over a n/w using cables.

• Cryptography

- It is used to secure & protect sensitive info by encoding it in a way that only authorized parties can understand.

- Encryption - Convert readable text (plaintext) to unreadable (ciphertext).

- Decryption: Convert ciphertext to plaintext.

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• Fundamental N/w security Principle

- Confidentiality
- Integrity
- Access ctrl
- Authentication
- Non repudiation
- Availability

• Types of Cryptography

① Symmetric - key crypto. (Secret key)

- Use same secret key for both encryption & decryption.

② Asymmetric

- Uses pair of public & private key for encry & decrypt.

③ Hash fn

- Transforms data into a fixed-size o/p to verify integrity, without using keys.

④ Digital Signature

- Confirms the sender's identity & msg authenticity using private key.

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• Multiplexing

- It is a tech. that combines multiple signals into one signal to transmit data more efficiently over a shared medium.

• Frequency Division Multiplexing (FDM)

- In this, a no. of signals are transmitted at same time, & each signal transmits its signals in the allocated frequency range.

- Appn : ① used in 1st generⁿ of mobile

② television broadcasting, FM, AM radio freq.

• Time Division Multiplexing (TDM)

- TDM is method that allows multiple signals to share a single commⁿ channel by dividing it into time slots so that each signal can use the channel.

Appn : digital TV

Date : _____

Types

- (i) Synchronous: fixed time slots
- (ii) Asynchronous: not fixed time slots
- (iii) Statistical: not fixed but depending on how much data transmitted

• Wavelength Division Multiplexing (WDM)

- It allows multiple data stream, to be transmitted simultaneously over a single fibre optic cable
- It used to increase bandwidth & optimize n/w infrastructure.

• Code Division Multi. (CDM)

- A tech. that allows multiple channels to share same freq. spectrum
- digital television, radio, mobile n/w.

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• Data Link Layer

- It is responsible for node-to-node delivery of msg

• Sub layers:

- Logical link control: It deals with protocol, flow-control, & error ctrl. and actual control of media.
- Media access control: It deals with actual control of media.

• functionality:

- (i) Framing
 - All takes the packets from n/w layer & encapsulates them into frames.
 - If frame size becomes too large, then packet divided into small sized frames.
- It control flow & error control

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② Addressing

- It uses Media Access Control (MAC) addresses to identify devices on a network.

- This process called physical addressing.

③ Synchronization

- When data frames are sent on the link, both ends must be synchronized in order to transfer to take place.

④ Error Control

① Flow Control

② Multi - Access

• Error Correction & Detection

• Errors

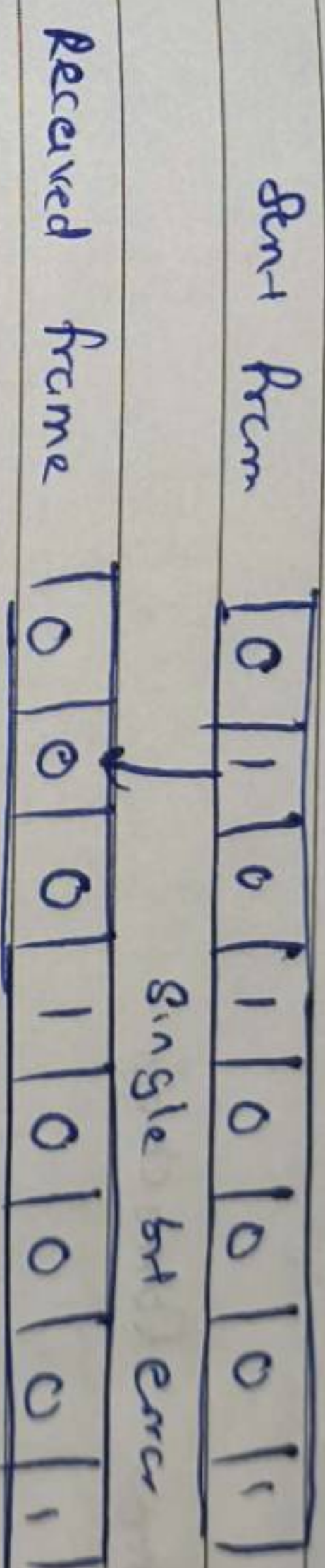
When bits are transmitted over the computer network, they are subject to get corrupted due to interference & network problems.

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• Types of Errors

① Single bit error

- In the received frame, only one bit has been corrupted.

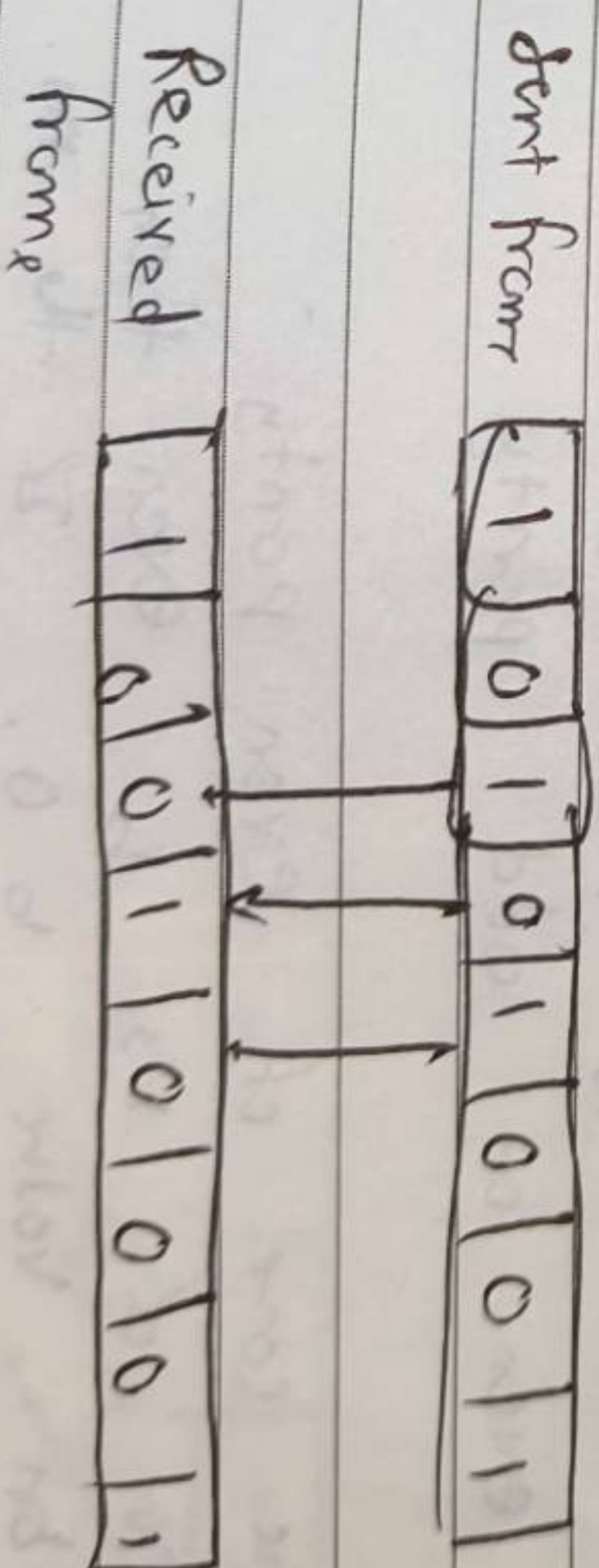


② Multiple bits error

- In the received frame, more than one bits are corrupted.

③ Burst error

- In the received frame, more than one consecutive bits are corrupted.



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• Error Control

• Error Detection

- It involves checking whether any error has occurred or not.

• Error Correction

- In this detected error has been solved

• Error Detection Techniques

• Parity Check

- The parity check is done by adding an extra bit, called parity bit, to the data to make a no. of 1's either even or odd parity.

- In the case of even parity :

If a no. of 1's is even then parity bit value is 0. If the no.

of 1's is odd then parity bit value is 1

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Data bit

1 1 0 1 0 1 1 1

Parity bit

= 1 = 5

1 1 1 1 0 0 0 1 0

= 1 = 4

- In the case of odd parity :

If the no. of 1's is odd then parity bit value is 0.

1 1 1 1 0 0 1 1 0

1 = 5

1 0 1 1 1 0 1 0 1

1 = 6

• Checksum

Sender

eg.

9 2 4 5 1 1 0 1

CS

- 21

Receiver

9 2 4 5 1 1 0

checksum (CS)

No error

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$$x_{CR} = \begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}$$

- Cyclic Redundancy Check (CRC)

Data - 1100

$$\text{Generator} = x^3 + x + 1 = 1011$$

Sender

1011) 1100000

1011

0111000

1011

010100

1011

00010

1100010

Sender

Receiver

1011) 1100010

1011

0111010

1011

010110

$$\text{Remainder} = 1011 = \boxed{4-1=3}$$

Date: _____

010110

1011

00000

→ 0

No error

- Error Correction Technique

- Forward Error Correction

- In this, the receiving end is responsible for correcting the n/w error.

- Backward Error Correction (Retransmission)

- In this, the sender is responsible for retransmission if data is error or detected by receiver.
- The receiver signals the sender to resend the corrupted data or the entire msg to ensure accurate delivery.

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Four main error correction codes

- Hamming codes
- Binary Convolution code
- Reed-Solomon code
- Low-Density Parity Check code

Hamming Code

- It is an error-correcting code used to ensure data accuracy during transmission & storage

Features:

- ① Error detection & correction
- ② Redundancy
- ③ Efficiency
- ④ Single error detection

Adv

- ① It is used in various fields such as comp. memory (RAM), common system data storage system.

disadv

- correct single bit errors
- It is not effective

Date: _____

- If Received H.C. is 1110101 with even parity then detect & correct error

$$P_1 = 2^0 = 1$$

$$P_2 = 2^1 = 2$$

$$P_4 = 2^2 = 4$$

D ₇	D ₆	D ₅	D ₄	D ₃	P ₂	P ₁
1	0	0	1	0	1	1

$$P_1 = D_3 \oplus D_5 \oplus D_7$$

$$1 = 1 \oplus 1 \oplus 1$$

$$P_1 = 0 \quad (\text{check 1, skip})$$

$$P_2 = D_3 \oplus D_6 \oplus D_7$$

$$0 = 1 \oplus 1 \oplus 1$$

$$P_2 = 1 \quad X$$

$$P_4 = D_5 \oplus D_6 \oplus D_7$$

$$0 = 1 \oplus 1 \oplus 1$$

$$P_4 = 1 \quad X$$

$$D_4 P_2 P_1 = 110 = 6^{\text{th}} \text{ bit error}$$

E	0	1	1	0	0	0	0	0	0
---	---	---	---	---	---	---	---	---	---

P	1	1	1	1	0	1	1	0	1
---	---	---	---	---	---	---	---	---	---

$$P \oplus E$$

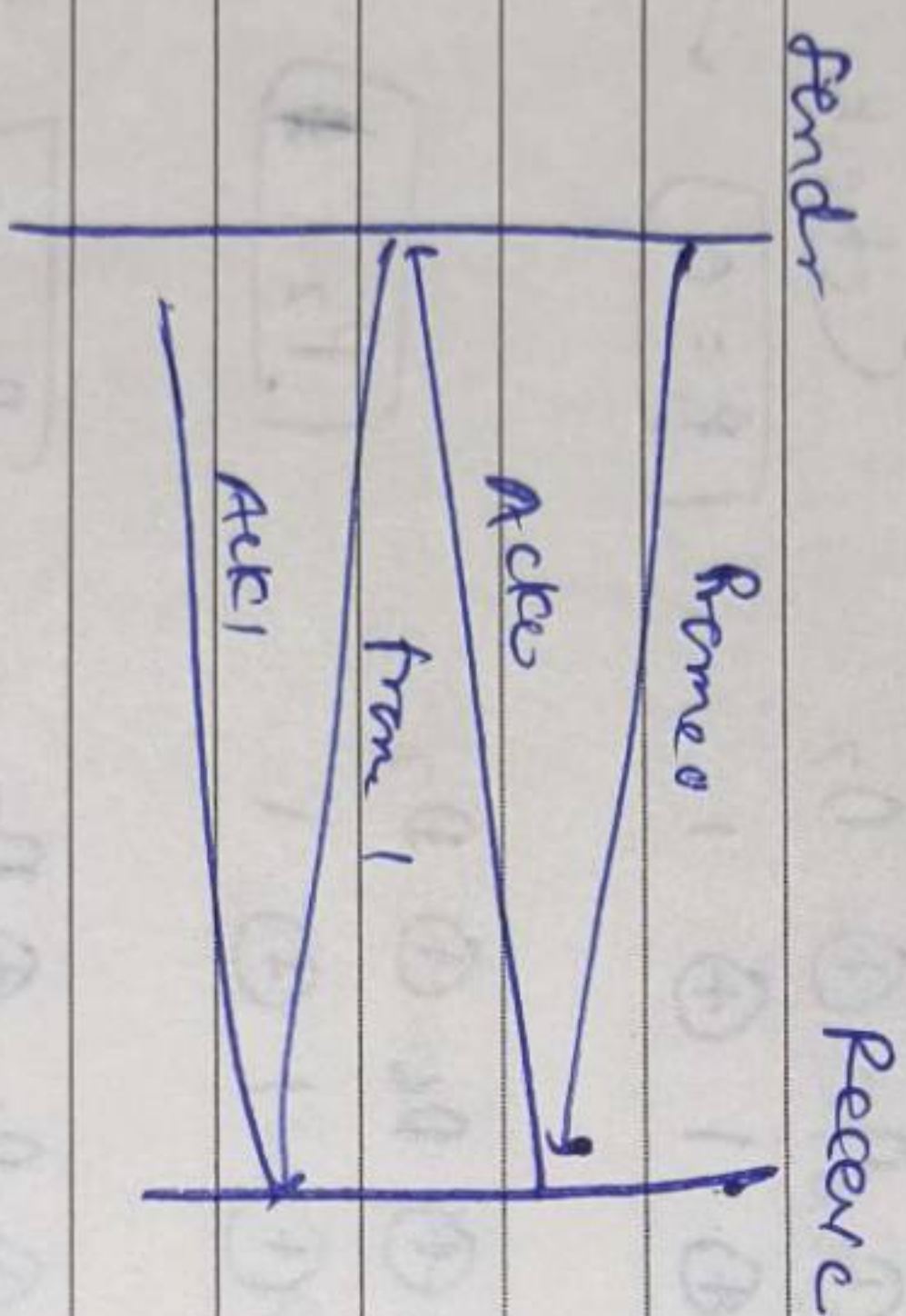
1	1	0	1	1	0	1	1	0	1
---	---	---	---	---	---	---	---	---	---

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* Data Link Control & Protocols

- Flow control : How & data from sender to receiver
- Stop & Wait .

- In this protocol , the sender sends a single data frame & then wait for an acknowledgment from receiver before sending the next frame



- Sliding Window Protocol
- It is a method in comp. networking that manages the flow of data betⁿ two devices , ensuring ~~these~~ devices reliably & sequential delivery of data frames

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- Error Control : It is process of detecting or identifying & re-transmitting data frames that might be lost or corrupted during transmission.

- Stop & Wait ARQ

- It send the data frame & then waits for an ack. After receive

- Timeout : duratⁿ for which sender waits for Ack from receiver. If sender does not receive Ack within timeout period , it assumes that the frame was lost & corrupted & retransmit the frame.

- Go Back N ARQ (Automatic Repeat Request)

- It is sliding window protocol used in networking for reliable data transmission.

- Selective Repeat ARQ

- It is sliding window technique to ensure reliable data frame delivery, particularly in noisy or unreliable channels.

• Network Layer

- It is used to maintain the quality of data & programme from source to destination

• Functions

- ① Assigning logical Address,
- ② Packaging
- ③ Host-to-Host delivery
- ④ Forwarding
- ⑤ Routing

- Network addressing

- It is the system that assign unique identifier (IP addr. MAC addr.)

- Network routing

- It is process of selecting paths for data packets to travel from source to destination

- Unicast routing

- This type of info. transfer is used when there is a participation of single sender & a single recipient

- Broadcast routing

- Transmission, data or info when single source sends packets to every device or node on net

- Multicast routing

- It allows sender to transmit data to specific recipient simultaneously instead of send separately.

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Anycom occurs

- It sends request to geologically closest available node

Routing Algorithm

① Floodlight

- When packet received, the router sends it to all interfaces / node but except the one which it was received

② Shortest Path (Dijkstra's algo)

Internetworking

- Interconnⁿ of multiple, independent comp. n/w to allow commⁿ & data exchange between them

Tunneling

- It allows data to be transmitted betⁿ two pts, effectively creating a private channel over public & shared n/w

Date : _____

New Layer Protocols

① ARP - Address Resolution Protocol

- It maps IP addr to MAC addr within local area n/w

② ICMP (Internet Control Msg Protocol)

- used to report errors & perform n/w diagnostics in IP n/w

③ IPv4 (32 bit)

- route data packets across n/w

④ IPv6 (128-bit)

⑤ RARP (Reverse)

⑥ IDPec

Adv.

① Reliable

② Router

③ Layered arch.

④ Scalability

Date : _____

• Disadv.

① Complexity

② Performance issue

③ Security

④ Dependency on infrastructure

• Routing

Modularity

• Routing table is required

• not required

• Traffic is less

• make

• Duplicate packets can not prevent

• Always give

• Always give

• Always give

per

Date : _____

• Application layer

• The appⁿ layer provides the functionality to send & receive data from user

- It acts as interface b/w user & appⁿ

• Pⁿ

① Provides services to appⁿ

② Data exchange

③ Protocol implementation

④ User interface

⑤ Data presentation & security

• Client - Server Model

① Peer-to-peer

② Client-server

Date : _____

• Communication

- Two Process in client-server model

① Socket

- It is interface that allow app to communicate with each other over the net

- They act as bridge b/w the app & transport layer.

② Remote Procedure call

- It enables program on one computer to execute a procedure on another computer in net.

• Application Protocols

① HTTP (Hypertext transfer Protocol) :

used for web browsing,

② SMTP : used to sending & receiving emails

Date : _____

③ POP : used to transferring files b/w comp.

④ DNS (Domain Name System) : used for translating domain names into IP address domain name (like google.com)

⑤ SNMP (Simple network management Protocol) : used to monitor & maintain net devices like router, switches, server & printers.

⑥ POP (Post office Protocol) : used to retrieve email from mail server.

• Network services

① Directory services

- Accounting : It stores info in specific form & makes available when requested

Date : _____

• Authentication & Authorization

- DNS (Domain Name Services)
- It turns domain name into IP address, which browser use to locate internet pages

⑪ File Services

- File sharing
- file transfer

⑫ Communication Services

- Email
- Social media
- Internet chat - Remote Access

⑬ Application Services

- Resource sharing
- Databases
- Web services

Date : _____

• Transport Layer :

- It offers peer-to-peer & end-to-end communication between two processes

- It takes data from upper layers & then breaks it into smaller size segments

• Functions

- ① The Process to Process Delivery.
- ② End-to-End Communication between hosts.
- ③ Multiplexing & Demultiplexing
- ④ Error Correction
- ⑤ Flow Control

• Transmission Control Protocol (TCP)

- It is connection-oriented protocol.
- It establishes secured communication between sender & receiver

Date : _____

• Features

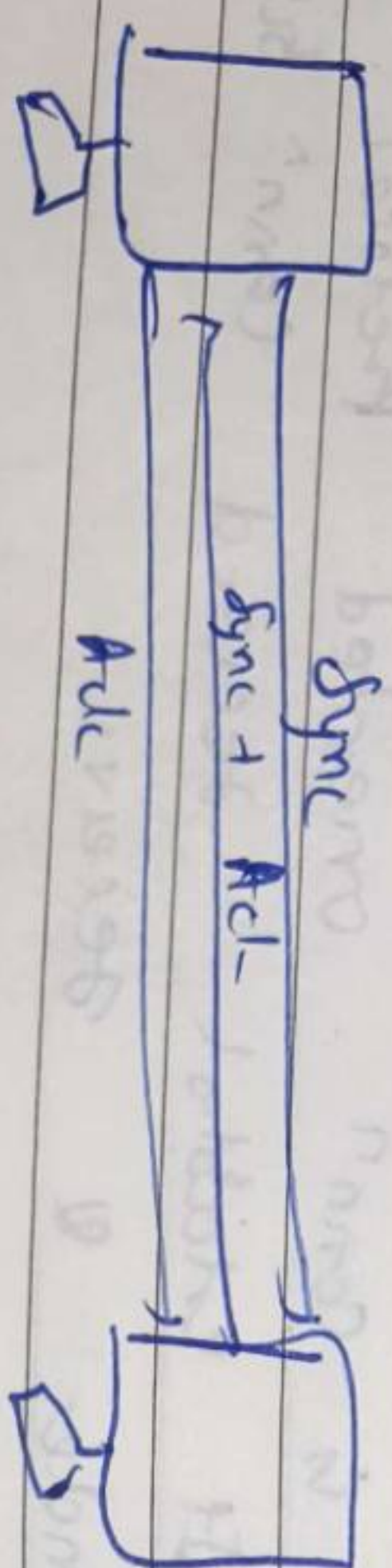
- ① Reliable
- ② Connⁿ oriented
- ③ end-to-end connⁿ
- ④ Point-to-point med

• Header

- The length of TCP header min: 20 bytes & max. 60 bytes.

Source Port		Destinat ⁿ Port	
Seq. no.			
Ack No.		Window size	
data offset	Reserved	Flags	Urgent
Checksum		Options	

- Diagram Desc. See docs Google



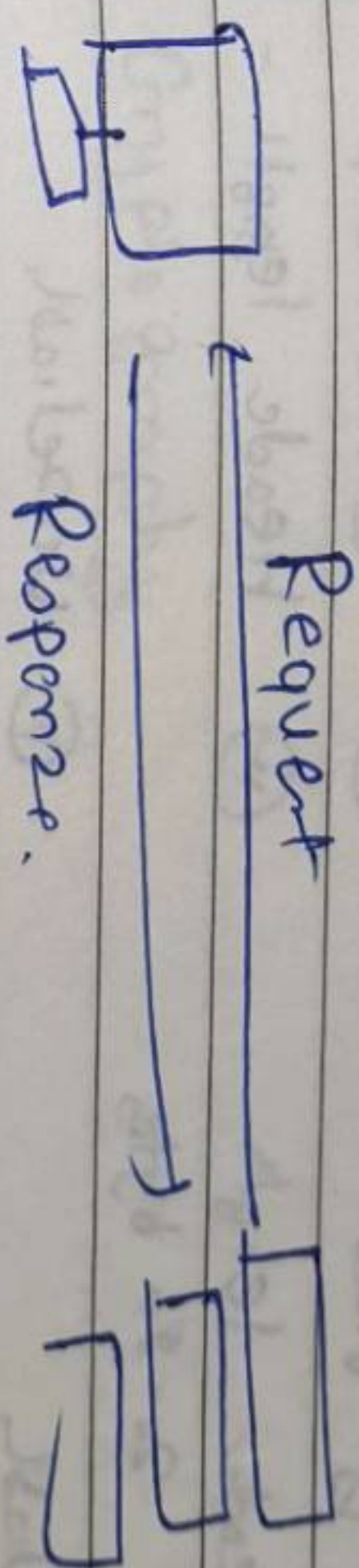
Working

Date : _____

• User Datagram Protocol (UDP)

- It is unreliable & Connⁿ less protocol.

• Working



- UDP Header - 8 Bytes.

UDP Header		UDP Data	
Source Port		Destination Port	
16 bits		16 bits	
Length		Checksum	
16 bits		16 bits	

Desc - in Google doc

Date : _____

TCP

UDP

(i) Connⁿ - oriented

(i) Connⁿ less, Protocol

Protocol

(ii) Data Delivery is guaranteed

(ii) not guaranteed

(iii) It is slow

(iii) It is fast

(iv) header length - 20-80 bytes

(iv) Header length - 8 bytes

(v) Reliable

(v) Unreliable

(vi) Ordered data transfer

(vi) Unordered

(vii) Error checking & error recovery process takes place

(viii) Error checking & error recovery process takes place

• Multiplexing

- The tech. to combine two or more data streams in one session

Date : _____

• Computer NW security

(i) Interruption

(ii) Privacy - Breach

(iii) Integrity

(iv) Authenticity

• Cryptography

- It is a tech. to encrypt the plain-text data which makes it difficult to understand & interpret

Symmetric

(i) Secret Key Encryption

- It uses a single shared key for both encryption & decryption data

Asymmetric

(ii) Private Key Encryption

- Two different keys used Private & Public

• Analog to Analog Conversion

• Digital to Analog Conversion

(Analog Transmission)

Basic Networking Interview Questions

1. What is an IPv4 address? What are the different classes of IPv4?

An IP address is a 32-bit dynamic address of a node in the network. An IPv4 address has 4 octets of 8-bit each with each number with a value up to 255.

IPv4 classes are differentiated based on the number of hosts it supports on the network. There are five types of IPv4 classes and are based on the first octet of IP addresses which are classified as Class A, B, C, D, or E.

IPv4 Class	IPv4 Start Address	IPv4 End Address	Usage
A	0.0.0.0	127.255.255.255	Used for Large Network
B	128.0.0.0	191.255.255.255	Used for Medium Size Network
C	192.0.0.0	223.255.255.255	Used for Local Area Network
D	224.0.0.0	239.255.255.255	Reserved for Multicasting
E	240.0.0.0	255.255.255.254	Study and R&D

Also, check out Scaler topics' Free [Computer Networks](#) course with certification to learn the fundamentals of computer networking.

2. Explain different types of networks.

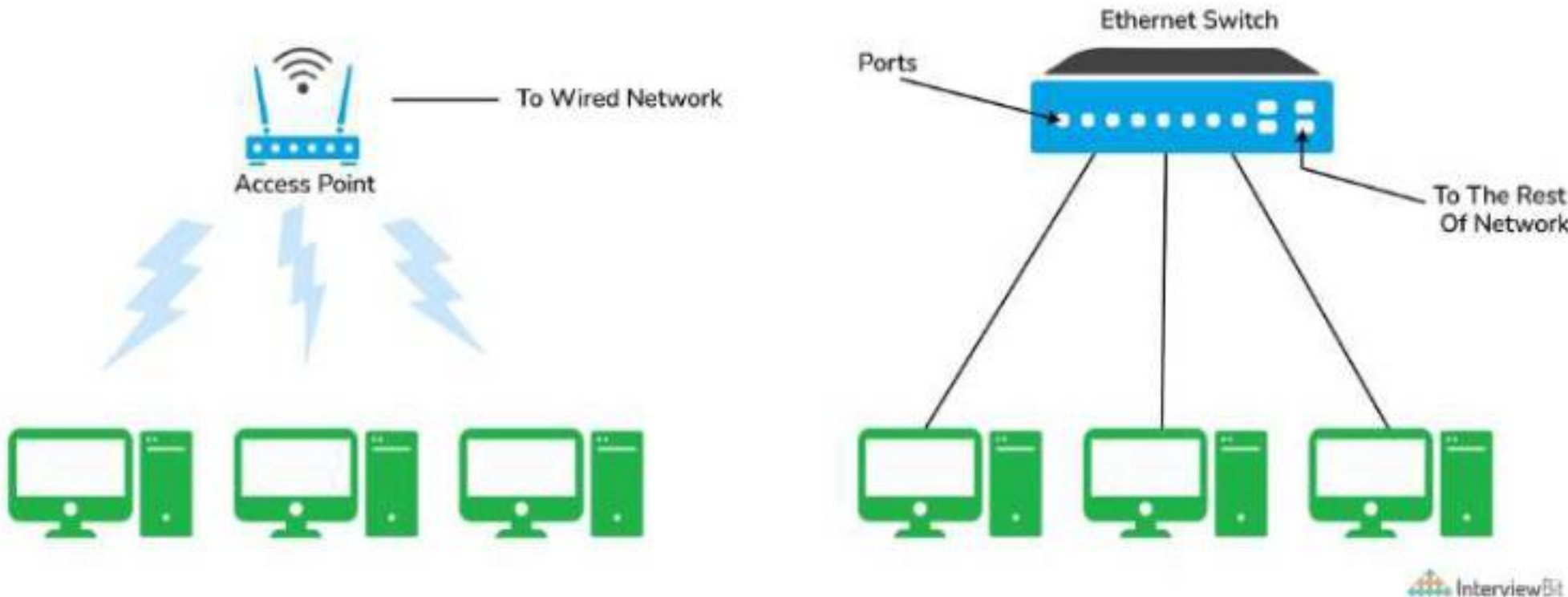
Below are few types of networks:

Type	Description
PAN (Personal Area Network)	Let devices connect and communicate over the range of a person. E.g. connecting Bluetooth devices.
LAN (Local Area Network)	It is a privately owned network that operates within and nearby a single building like a home, office, or factory
MAN (Metropolitan Area Network)	It connects and covers the whole city. E.g. TV Cable connection over the city
WAN (Wide Area Network)	It spans a large geographical area, often a country or continent. The Internet is the largest WAN
GAN (Global Area Network)	It is also known as the Internet which connects the globe using satellites. The Internet is also called the Network of WANs.

3. Explain LAN (Local Area Network)

LANs are widely used to connect computers/laptops and consumer electronics which enables them to share resources (e.g., printers, fax machines) and exchange information. When LANs are used by companies or organizations, they are called **enterprise networks**. There are two different types of LAN networks i.e. wireless LAN (no wires involved achieved using Wi-Fi) and wired LAN (achieved using LAN cable). Wireless LANs are very popular these days for places where installing wire is difficult. The below diagrams explain both wireless and wired LAN.

LAN (Local Area Network)



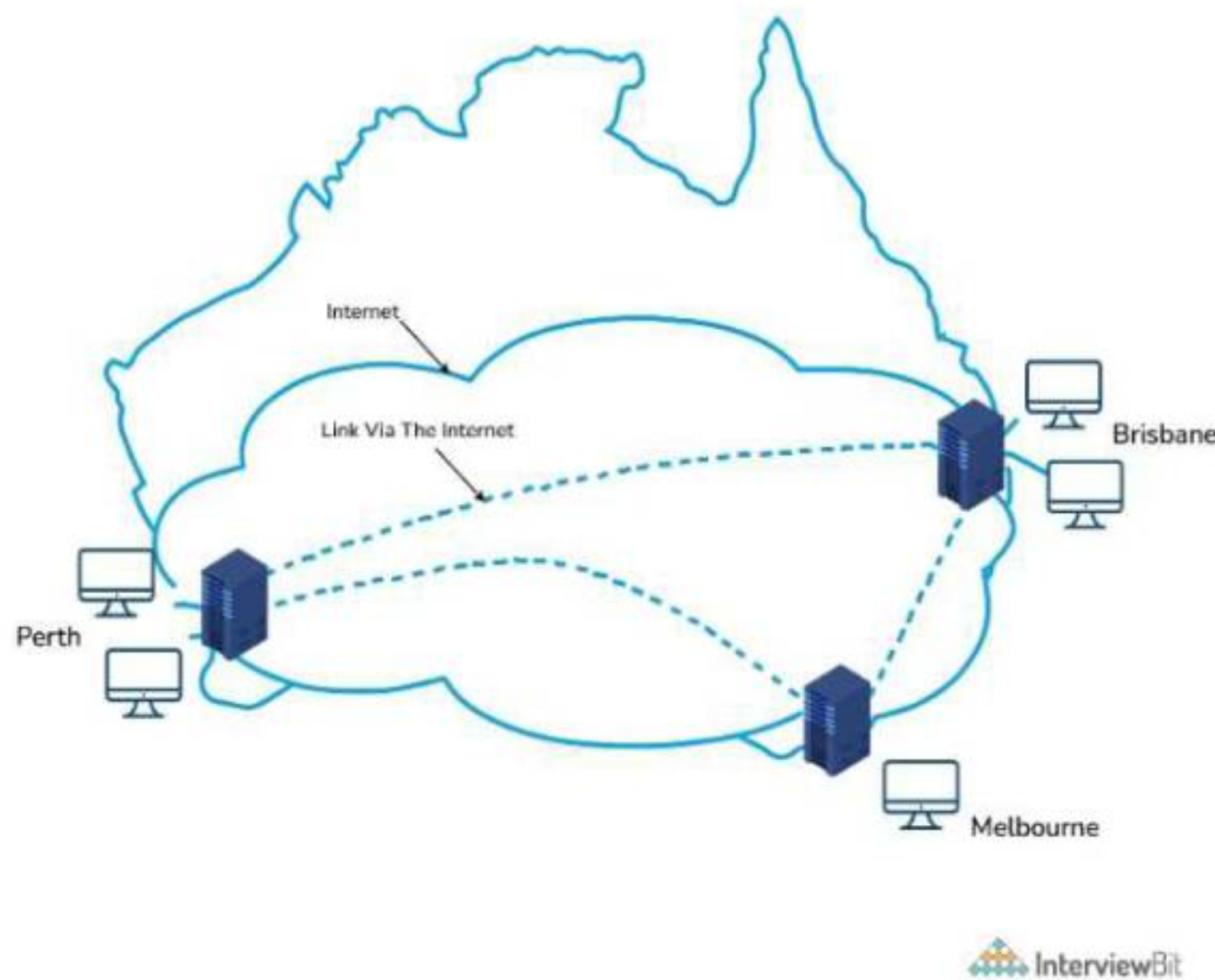
4. Tell me something about VPN (Virtual Private Network)

VPN or the Virtual Private Network is a private WAN (Wide Area Network) built on a public network (like the Internet) using a tunnel (protected network) between different networks using the internet (public network).

4. Tell me something about VPN (Virtual Private Network)

VPN or the Virtual Private Network is a private WAN (Wide Area Network) built on the internet. It allows the creation of a secured tunnel (protected network) between different networks using the internet (public network). By using the VPN, a client can connect to the organization's network remotely. The below diagram shows an organizational WAN network over Australia created using VPN:

VPN (Vir



5. What are the advantages of using a VPN?

Below are few advantages of using VPN:

- VPN is used to connect offices in different geographical locations remotely and is cheaper when compared to WAN connections.
- VPN is used for secure transactions and confidential data transfer between multiple offices located in different geographical locations.
- VPN keeps an organization's information secured against any potential threats or intrusions by using virtualization.
- VPN encrypts the internet traffic and disguises the online identity.

6. What are the different types of VPN?

Few types of VPN are:

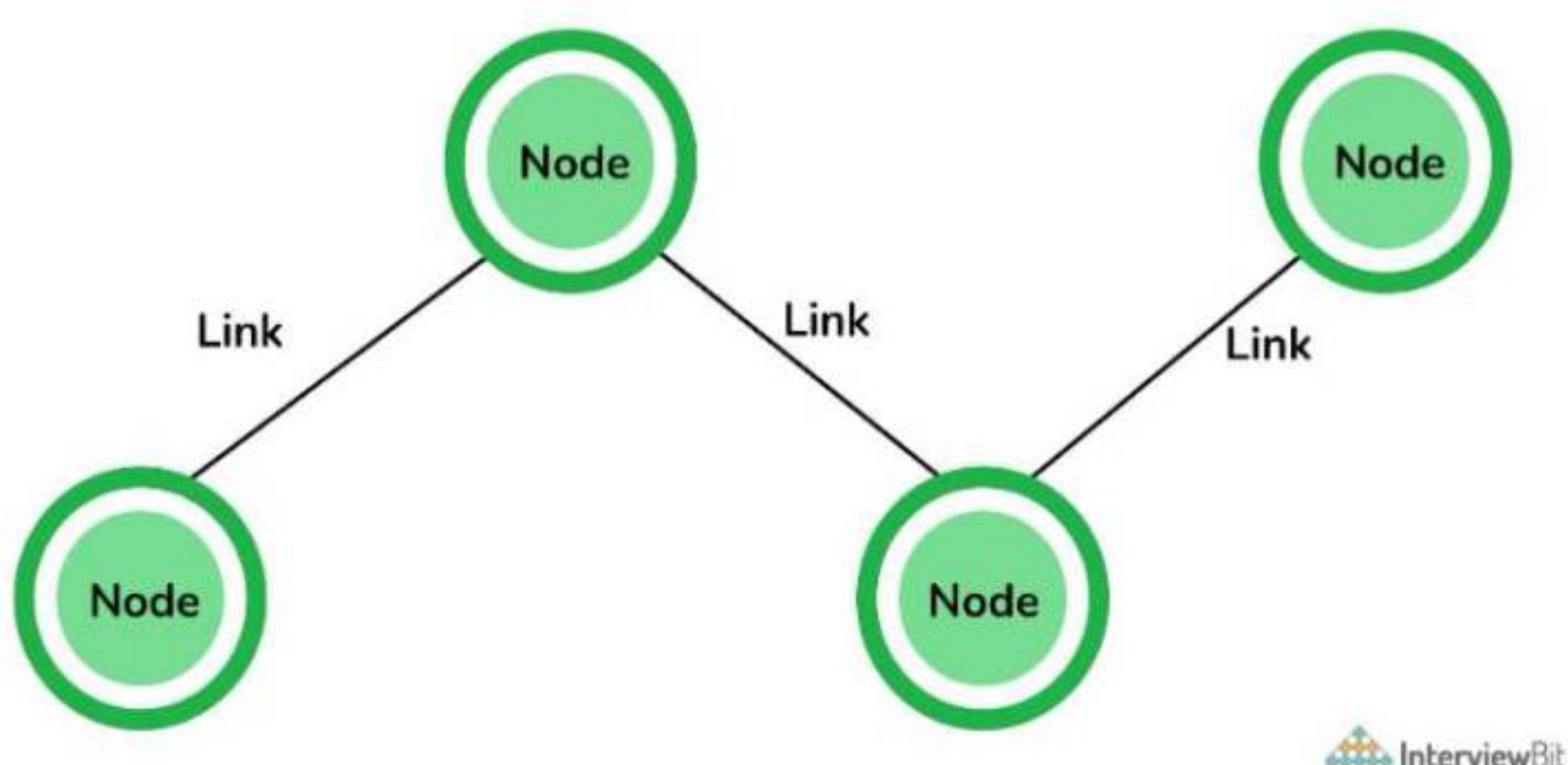
- **Access VPN:** Access VPN is used to provide connectivity to remote mobile users and telecommuters. It serves as an alternative to dial-up connections or ISDN (Integrated Services Digital Network) connections. It is a low-cost solution and provides a wide range of connectivity.
- **Site-to-Site VPN:** A Site-to-Site or Router-to-Router VPN is commonly used in large companies having branches in different locations to connect the network of one office to another in different locations. There are 2 sub-categories as mentioned below:
- **Intranet VPN:** Intranet VPN is useful for connecting remote offices in different geographical locations using shared infrastructure (internet connectivity and servers) with the same accessibility policies as a private WAN (wide area network).
- **Extranet VPN:** Extranet VPN uses shared infrastructure over an intranet, suppliers, customers, partners, and other entities and connects them using dedicated connections.

7. What are nodes and links?

Node: Any communicating device in a network is called a Node. Node is the point of intersection in a network. It can send/receive data and information within a network. Examples of the node can be computers, laptops, printers, servers, modems, etc.

Link: A link or edge refers to the connectivity between two nodes in the network. It includes the type of connectivity (wired or wireless) between the nodes and protocols used for one node to be able to communicate with the other.

Nodes and Links



8. What is the network topology?

8. What is the network topology?

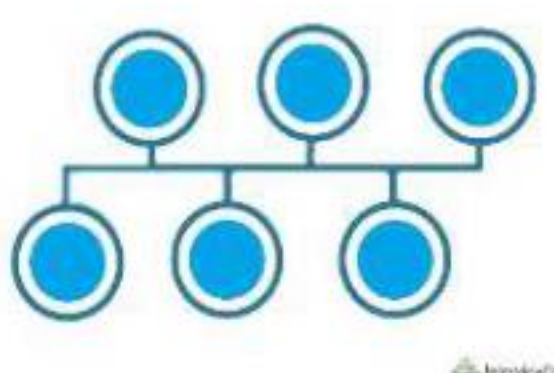
Network topology is a physical layout of the network, connecting the different nodes using the links. It depicts the connectivity between the computers, devices, cables, etc.

9. Define different types of network topology

The different types of network topology are given below:

Bus Topology:

Bus Topology



- All the nodes are connected using the central link known as the bus.
- It is useful to connect a smaller number of devices.
- If the main cable gets damaged, it will damage the whole network.

Star Topology:

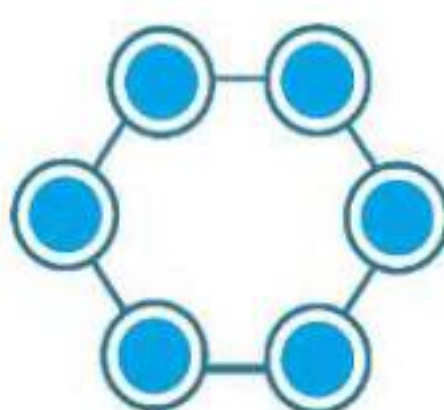
Star Topology



- All the nodes are connected to one single node known as the central node.
- It is more robust.
- If the central node fails the complete network is damaged.
- Easy to troubleshoot.
- Mainly used in home and office networks.

Ring Topology:

Ring Topology

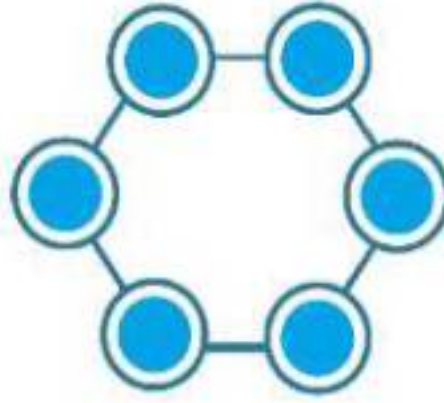


- Easy to troubleshoot.

- Mainly used in home and office networks.

Ring Topology:

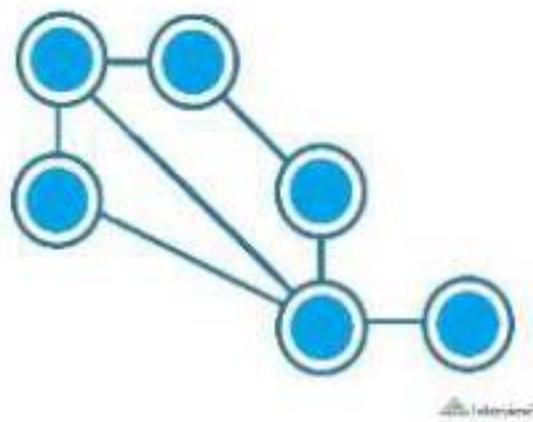
Ring Topology



- Each node is connected to exactly two nodes forming a ring structure
- If one of the nodes are damaged, it will damage the whole network
- It is used very rarely as it is expensive and hard to install and manage

Mesh Topology:

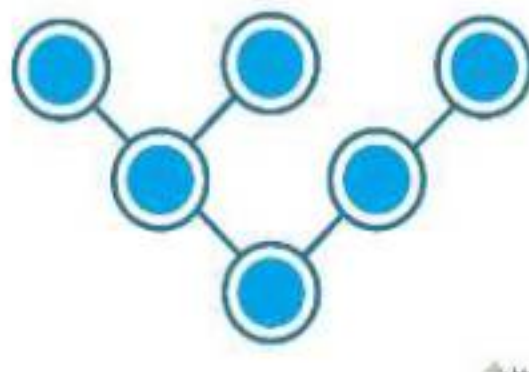
Mesh Topology



- Each node is connected to one or many nodes.
- It is robust as failure in one link only disconnects that node.
- It is rarely used and installation and management are difficult.

Tree Topology:

Tree Topology



- A combination of star and bus topology also know as an extended bus topology.
- All the smaller star networks are connected to a single bus.
- If the main bus fails, the whole network is damaged.

Hybrid:

- It is a combination of different topologies to form a new topology.
- It helps to ignore the drawback of a particular topology and helps to pick the

- All the smaller star networks are connected to a single bus.
- If the main bus fails, the whole network is damaged.

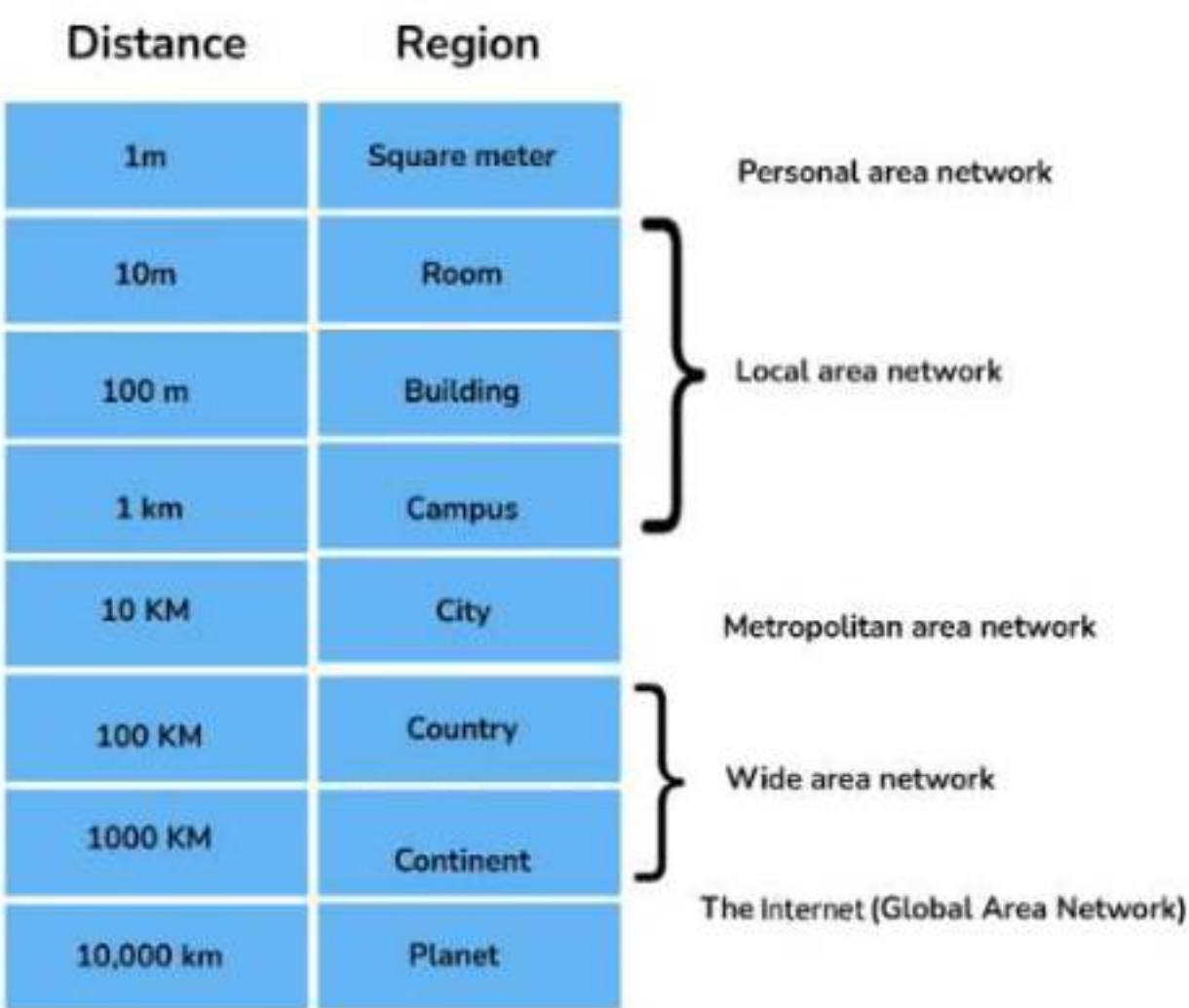
Hybrid:

- It is a combination of different topologies to form a new topology.
- It helps to ignore the drawback of a particular topology and helps to pick the strengths from other.

10. How are Network types classified?

Network types can be classified and divided based on the area of distribution of the network. The below diagram would help to understand the same:

Network Types



11. What are Private and Special IP addresses?

Private Address: For each class, there are specific IPs that are reserved specifically for private use only. This IP address cannot be used for devices on the Internet as they are non-routable.

IPv4 Class	Private IPv4 Start Address	Private IPv4 End Address
A	10.0.0.0	10.255.255.255
B	172.16.0.0	172.31.255.255
C	192.168.0.0	192.168.255.255

Special Address: IP Range from 127.0.0.1 to 127.255.255.255 are network testing addresses also known as loopback addresses are the special IP address.

Intermediate Interview Questions

12. What is the DNS?

DNS is the Domain Name System. It is considered as the devices/services directory of the Internet. It is a decentralized and hierarchical naming system for devices/services connected to the Internet. It translates the domain names to their corresponding IPs. For e.g. interviewbit.com to 172.217.166.36. It uses port 53 by default.

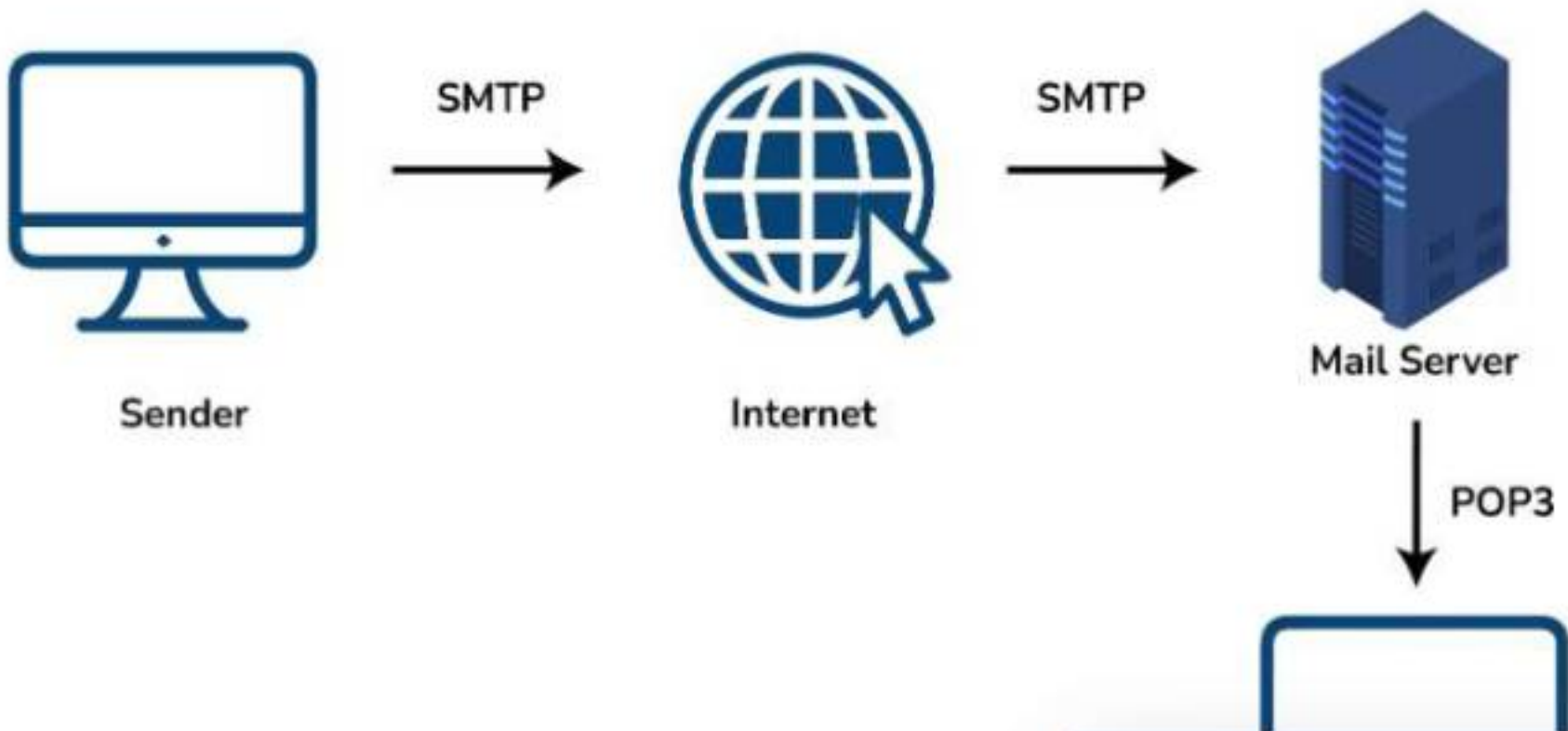
13. What is the use of a router and how is it different from a gateway?

The router is a networking device used for connecting two or more network segments. It directs the traffic in the network. It transfers information and data like web pages, emails, images, videos, etc. from source to destination in the form of packets. It operates at the network layer. The gateways are also used to route and regulate the network traffic but, they can also send data between two dissimilar networks while a router can only send data to similar networks.

14. What is the SMTP protocol?

SMTP is the Simple Mail Transfer Protocol. SMTP sets the rule for communication between servers. This set of rules helps the software to transmit emails over the internet. It supports both End-to-End and Store-and-Forward methods. It is in always-listening mode on port 25.

SMTP Protocol

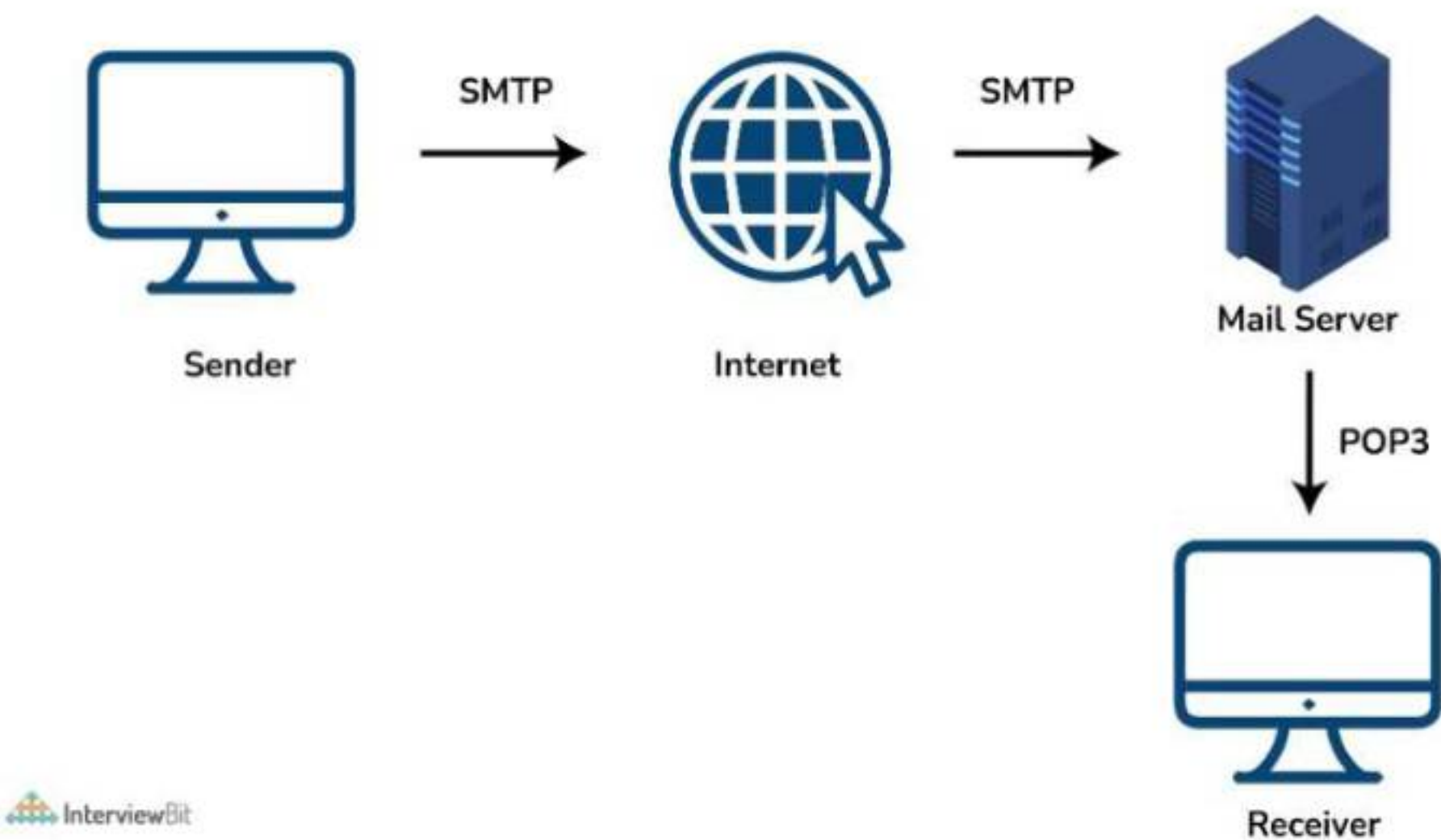


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SMTP Protocol



15. Describe the OSI Reference Model

Open System Interconnections (OSI) is a network architecture model based on the ISO standards. It is called the OSI model as it deals with connecting the systems that are open for communication with other systems.

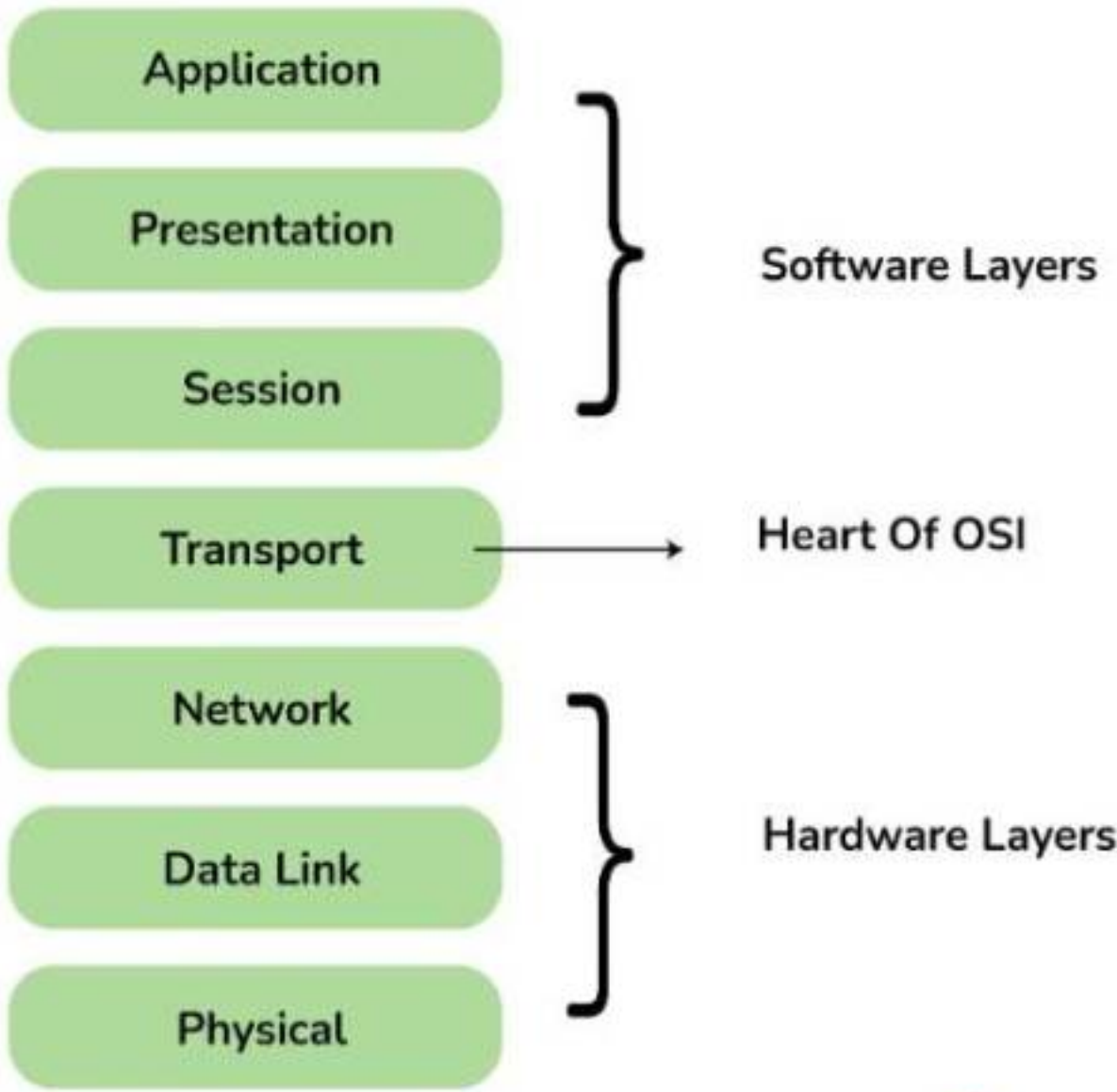
The OSI model has seven layers. The principles used to arrive at the seven layers can be summarized briefly as below:

- Create a new layer if a different abstraction is needed.
- Each layer should have a well-defined function.
- The function of each layer is chosen based on internationally standardized protocols.

16. Define the 7 different layers of the OSI Reference Model

Here the 7 layers of the OSI reference model:

Layers o



Layer	Unit Exchanged	Description
Physical	Bit	• It is concerned with transmitting raw bits over a communication channel. • Chooses which type of transmission mode is to be selected for the transmission. The available transmission modes are Simplex, Half Duplex and Full Duplex.,
Data Link	Frame	• The main task of this layer is to transform a raw transmission facility into a line that appears free of undetected transmission errors. • It also allows detecting damaged packets using the CRC (Cyclic Redundancy Check) error-detecting, code. • When more than one node is connected to a shared link, Data Link Layer protocols are required to determine which device has control over the link at a given time. • It is implemented by protocols like CSMA/CD, CSMA/CA, ALOHA, and Token Passing.

3:28

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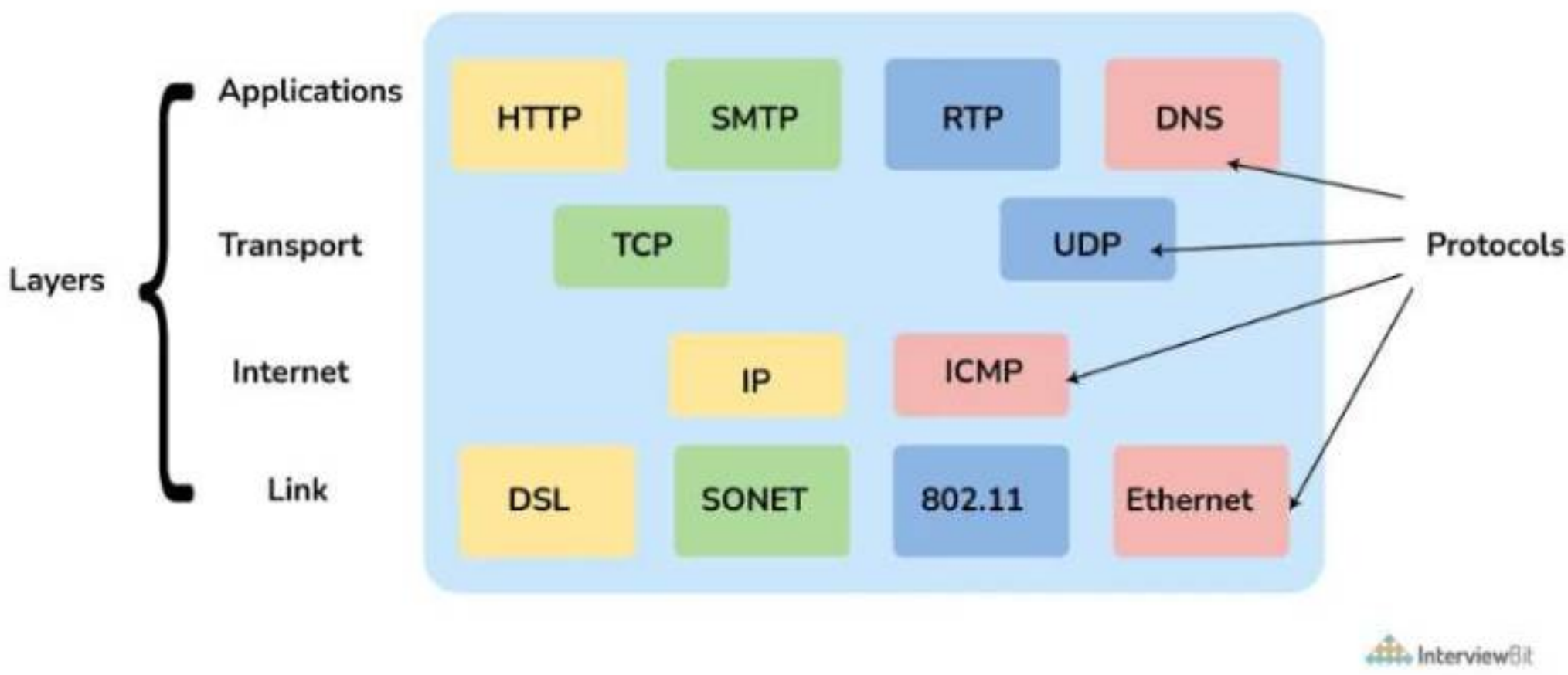
Layer	Unit Exchanged	Description
Physical	Bit	- It is concerned with transmitting raw bits over a communication channel. - Chooses which type of transmission mode is to be selected for the transmission. The available transmission modes are Simplex, Half Duplex and Full Duplex.,
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Network	Packet	- It controls the operation of the subnet. - The network layer takes care of feedback messaging through ICMP messages.
Transport	TPDU - Transaction Protocol Data Unit	- The basic functionality of this layer is to accept data from the above layers, split it up into smaller units if needed, pass these to the network layer, and ensure that all the pieces arrive correctly at the other end. - The Transport Layer takes care of Segmentation and Reassembly.
Session	SPDU - Session Protocol Data Unit	- The session layer allows users on different machines to establish sessions between them. - Dialogue control is using the full-duplex link as half-duplex. It sends out dummy packets from the client to the server when the client is ideal.
Presentation	PPDU - Presentation Protocol Data Unit	- The presentation layer is concerned with the syntax and semantics of the information transmitted. - It translates a message from a common form to the encoded format which will be understood by the receiver.
Application	APDU - Application Protocol Data Unit	- It contains a variety of protocols that are commonly needed by users. - The application layer sends data of any size to the transport layer.

17. Describe the TCP/IP Reference Model

It is a compressed version of the OSI model with only 4 layers. It was developed by the US Department of Defence (DoD) in the 1980s. The name of this model is based on 2 standard protocols used i.e. TCP (Transmission Control Protocol) and IP (Internet Protocol).

18. Define the 4 different layers of the TCP/IP Reference Model

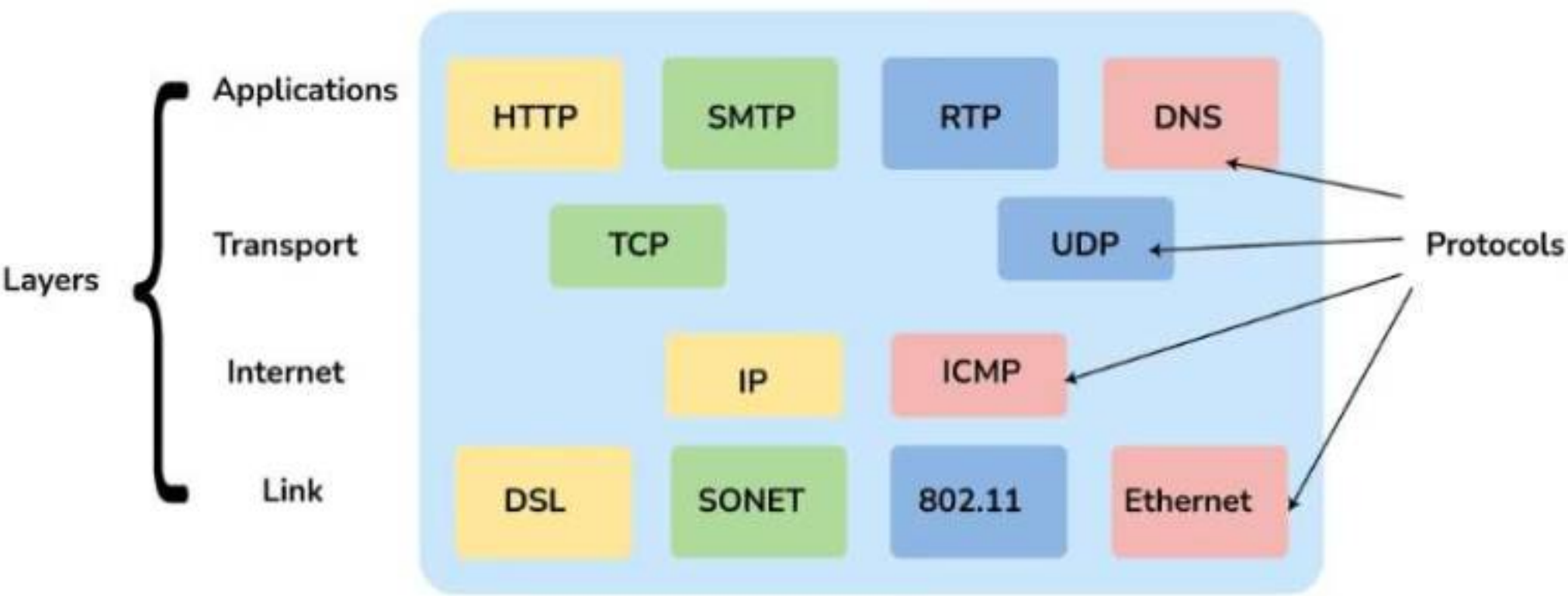
Layers of TCP/IP



Layer	Description
-------	-------------

18. Define the 4 different layers of the TCP/IP Reference Model

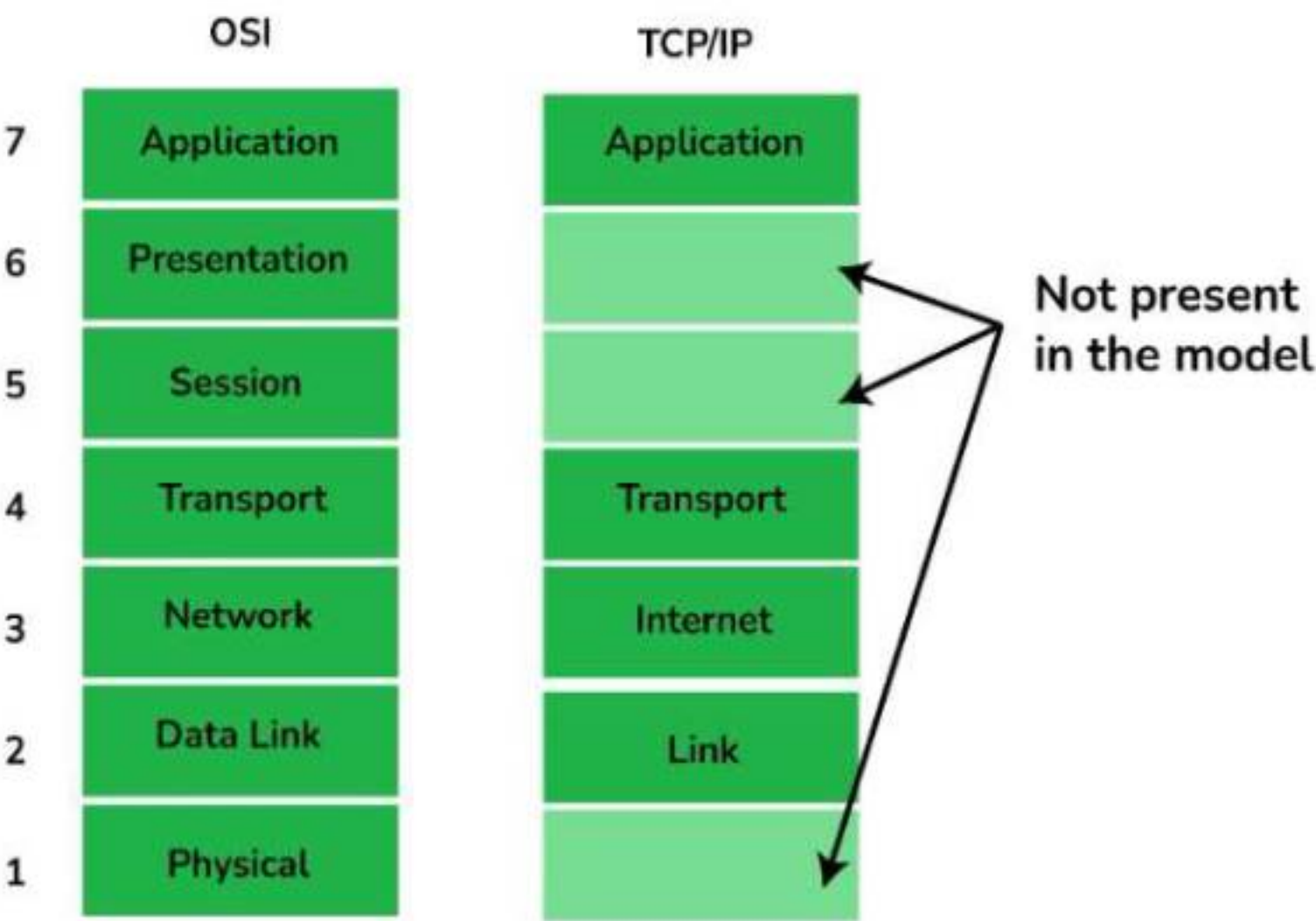
Layers of TCP/IP



Layer	Description
Link	Decides which links such as serial lines or classic Ethernet must be used to meet the needs of the connectionless internet layer.
Internet	- The internet layer is the most important layer which holds the whole architecture together. - It delivers the IP packets where they are supposed to be delivered.
Transport	Its functionality is almost the same as the OSI transport layer. It enables peer entities on the network to carry on a conversation.
Application	It contains all the higher-level protocols.

19. Differentiate OSI Reference Model with TCP/IP Reference Model

OSI Vs TCF



OSI Reference Model	TCP/IP Reference Model
7 layered architecture	4 layered architecture
Fixed boundaries and functionality for each layer	Flexible architecture with no strict boundaries between layers
Low Reliability	High Reliability
Vertical Layer Approach	Horizontal Layer Approach

20. What are the HTTP and the HTTPS protocol?

HTTP is the HyperText Transfer Protocol which defines the set of rules and standards on how the information can be transmitted on the World Wide Web (WWW). It helps the web browsers and web servers for communication. It is a 'stateless protocol' where each command is independent with respect to the previous command. HTTP is an application layer protocol built upon the TCP. It uses port 80 by default.

HTTPS is the HyperText Transfer Protocol Secure or Secure HTTP. It is an advanced and secured version of HTTP. On top of HTTP, SSL/TLS protocol is used to provide security. It enables secure transactions by encrypting the communication and also helps identify network servers securely. It uses port 443 by default.

Advanced Interview Questions

21. What is the FTP protocol?

FTP is a File Transfer Protocol. It is an application layer protocol used to transfer files from one computer to another. It can also be used to download files from remote servers to your computer. It uses two ports: 21 for control and 20 for data.

HTTPS is the HyperText Transfer Protocol Secure or Secure HTTP. It is an advanced and secured version of HTTP. On top of HTTP, SSL/TLS protocol is used to provide security. It enables secure transactions by encrypting the communication and also helps identify network servers securely. It uses port 443 by default.

Advanced Interview Questions

21. What is the FTP protocol?

FTP is a File Transfer Protocol. It is an application layer protocol used to transfer files and data reliably and efficiently between hosts. It can also be used to download files from remote servers to your computer. It uses port 27 by default.

22. What is the TCP protocol?

TCP or TCP/IP is the Transmission Control Protocol/Internet Protocol. It is a set of rules that decides how a computer connects to the Internet and how to transmit the data over the network. It creates a virtual network when more than one computer is connected to the network and uses the three ways handshake model to establish the connection which makes it more reliable.

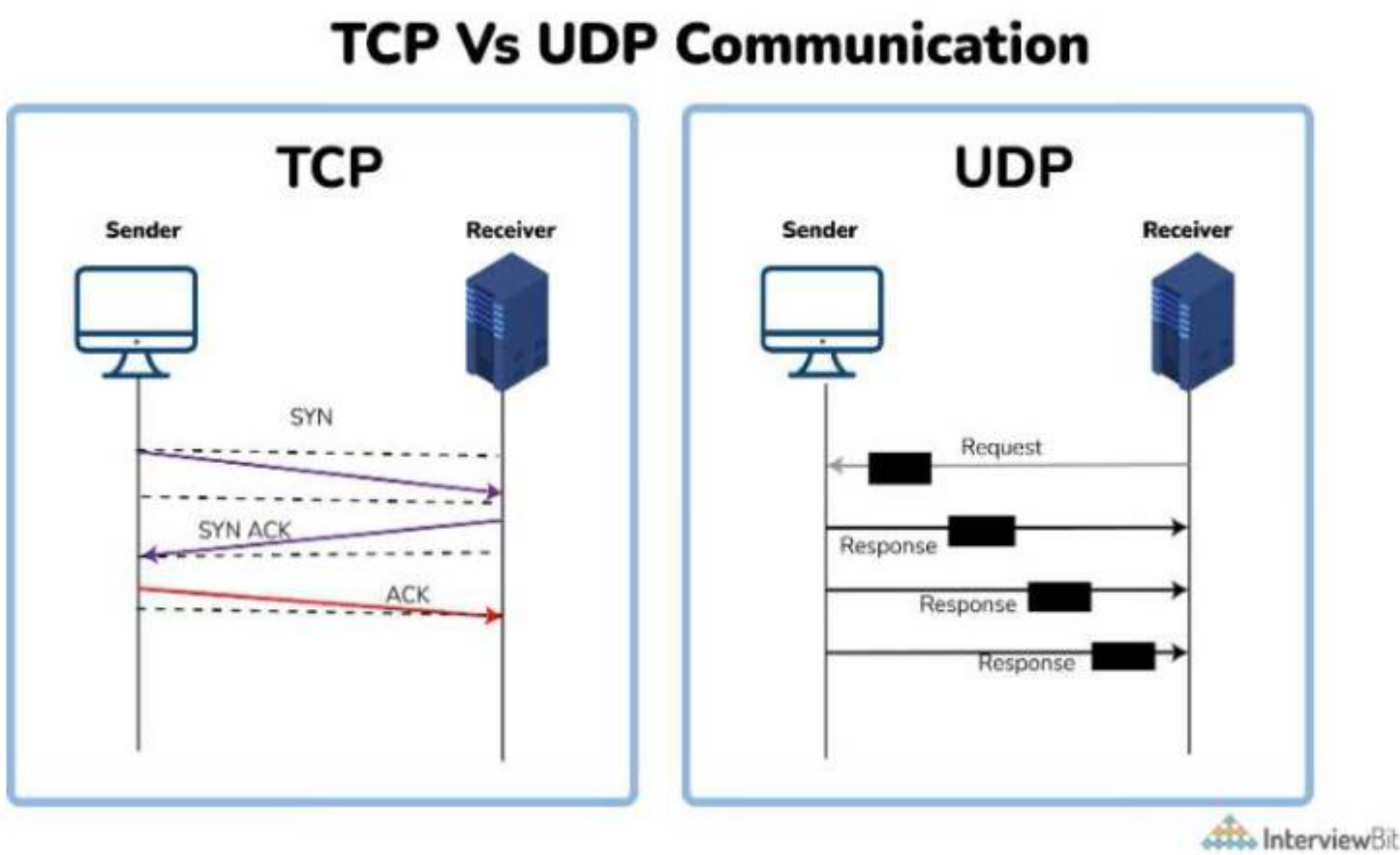
23. What is the UDP protocol?

UDP is the User Datagram Protocol and is based on Datagrams. Mainly, it is used for multicasting and broadcasting. Its functionality is almost the same as TCP/IP Protocol except for the three ways of handshaking and error checking. It uses a simple transmission without any hand-shaking which makes it less reliable.

24. Compare between TCP and UDP

TCP/IP	UDP
Connection-Oriented Protocol	Connectionless Protocol
More Reliable	Less Reliable
Slower Transmission	Faster Transmission
Packets order can be preserved or can be rearranged	Packets order is not fixed and packets are independent of each other
Uses three ways handshake model for connection	No handshake for establishing the connection
TCP packets are heavy-weight	UDP packets are light-weight
Offers error checking mechanism	No error checking mechanism
Protocols like HTTP, FTP, Telnet, SMTP, HTTPS, etc use TCP at the transport layer	Protocols like DNS, RIP, SNMP, RTP, BOOTP, TFTP, NIP, etc use UDP at the transport layer

TCP VS UDP



25. What is the ICMP protocol?

ICMP is the Internet Control Message Protocol. It is a network layer protocol used for error handling. It is mainly used by network devices like routers for diagnosing the network connection issues and crucial for error reporting and testing if the data is reaching the preferred destination in time. It uses port 7 by default.

26. What do you mean by the DHCP Protocol?

DHCP is the Dynamic Host Configuration Protocol.

It is an application layer protocol used to auto-configure devices on IP networks enabling them to use the TCP and UDP-based protocols. The DHCP servers auto-assign the IPs and other network configurations to the devices individually which enables them to communicate over the IP network. It helps to get the subnet mask, IP address and helps to resolve the DNS. It uses port 67 by default.

27. What is the ARP protocol?

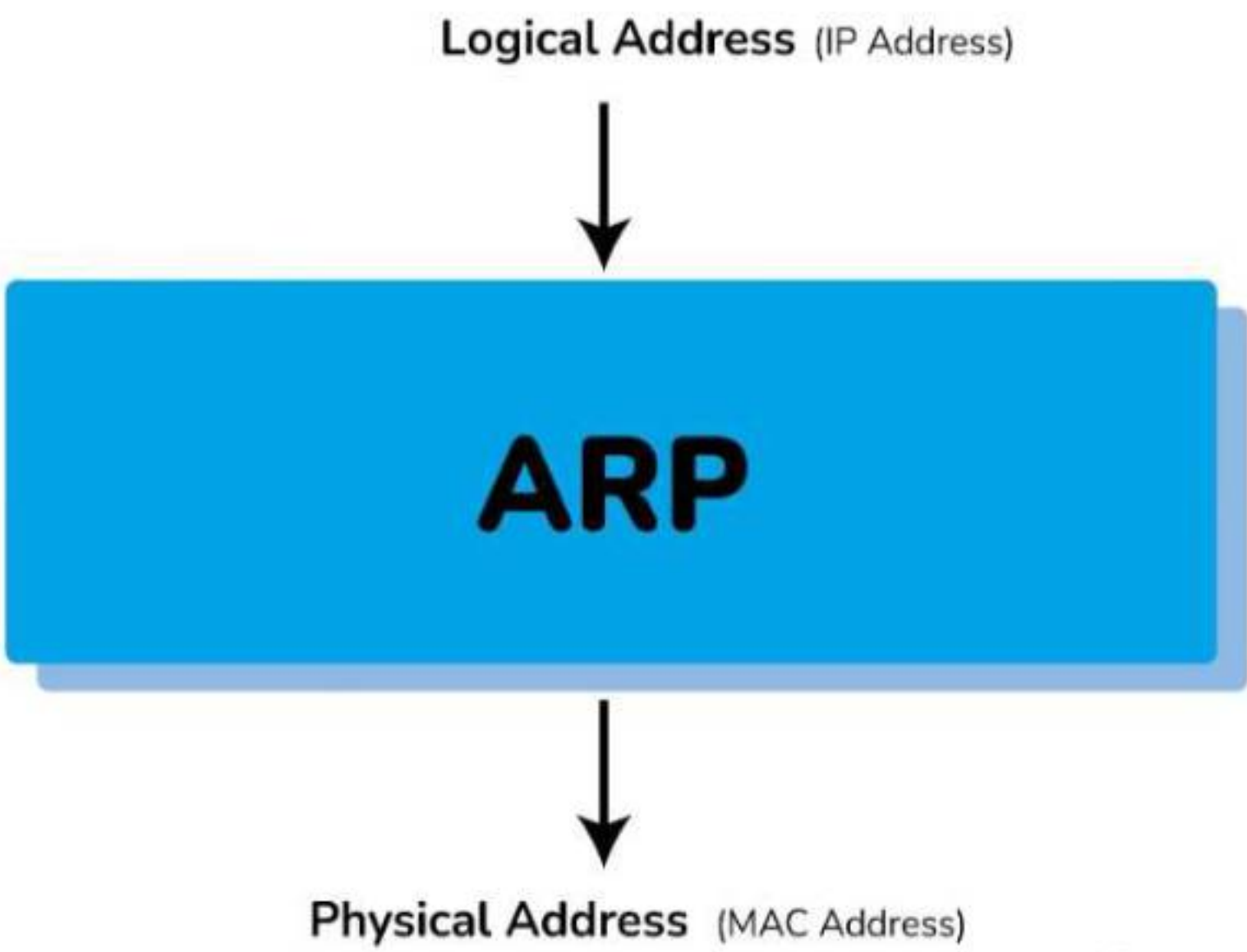
ARP is Address Resolution Protocol. It is a network-level protocol used to convert the logical address i.e. IP address to the device's physical address i.e. MAC address. It can also be used to get the MAC address of devices when they are trying to communicate over the local network.

It is an application layer protocol used to auto-configure devices on IP networks enabling them to use the TCP and UDP-based protocols. The DHCP servers auto-assign the IPs and other network configurations to the devices individually which enables them to communicate over the IP network. It helps to get the subnet mask, IP address and helps to resolve the DNS. It uses port 67 by default.

27. What is the ARP protocol?

ARP is Address Resolution Protocol. It is a network-level protocol used to convert the logical address i.e. IP address to the device's physical address i.e. MAC address. It can also be used to get the MAC address of devices when they are trying to communicate over the local network.

ARP Pr



28. What is the MAC address and how is it related to NIC?

MAC address is the Media Access Control address. It is a 48-bit or 64-bit unique identifier of devices in the network. It is also called the physical address embedded with Network Interface Card (NIC) used at the Data Link Layer. NIC is a hardware component in the networking device using which a device can connect to the network.

29. Differentiate the MAC address with the IP address

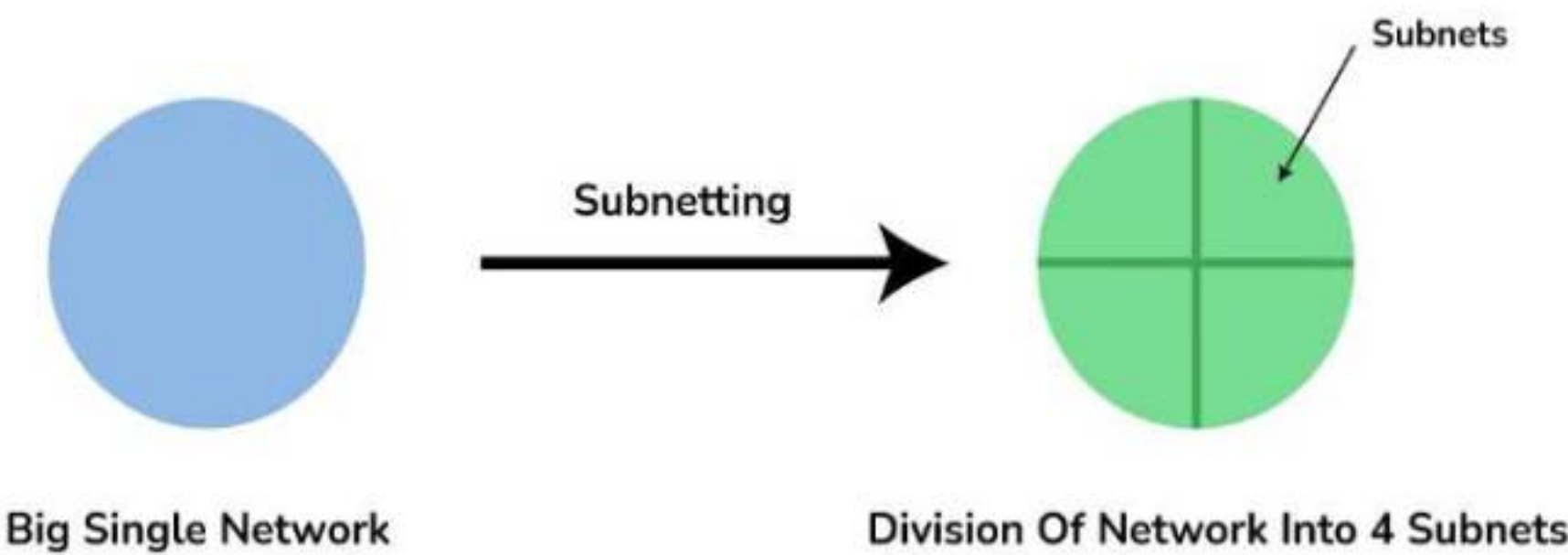
The difference between MAC address and IP address are as follows:

MAC Address	IP Address
Media Access Control Address	Internet Protocol Address
6 or 8-byte hexadecimal number	4 (IPv4) or 16 (IPv6) Byte address
It is embedded with NIC	It is obtained from the network
Physical Address	Logical Address
Operates at Data Link Layer	Operates at Network Layer.
Helps to identify the device	Helps to identify the device connectivity on the network.

30. What is a subnet?

A subnet is a network inside a network achieved by the process called subnetting which helps divide a network into subnets. It is used for getting a higher routing efficiency and enhances the security of the network. It reduces the time to extract the host address from the routing table.

Subnet



31. Compare the hub vs switch

Hub	Switch
Operates at Physical Layer	Operates at Data Link Layer

31. Compare the hub vs switch

Hub	Switch
Operates at Physical Layer	Operates at Data Link Layer
Half-Duplex transmission mode	Full-Duplex transmission mode
Ethernet devices can be connected	LAN devices can be connected
Less complex, less intelligent, and cheaper	Intelligent and effective
No software support for the administration	Administration software support is present
Less speed up to 100 MBPS	Supports high speed in GBPS
Less efficient as there is no way to avoid collisions when more than one nodes sends the packets at the same time	More efficient as the collisions can be avoided or reduced as compared to Hub

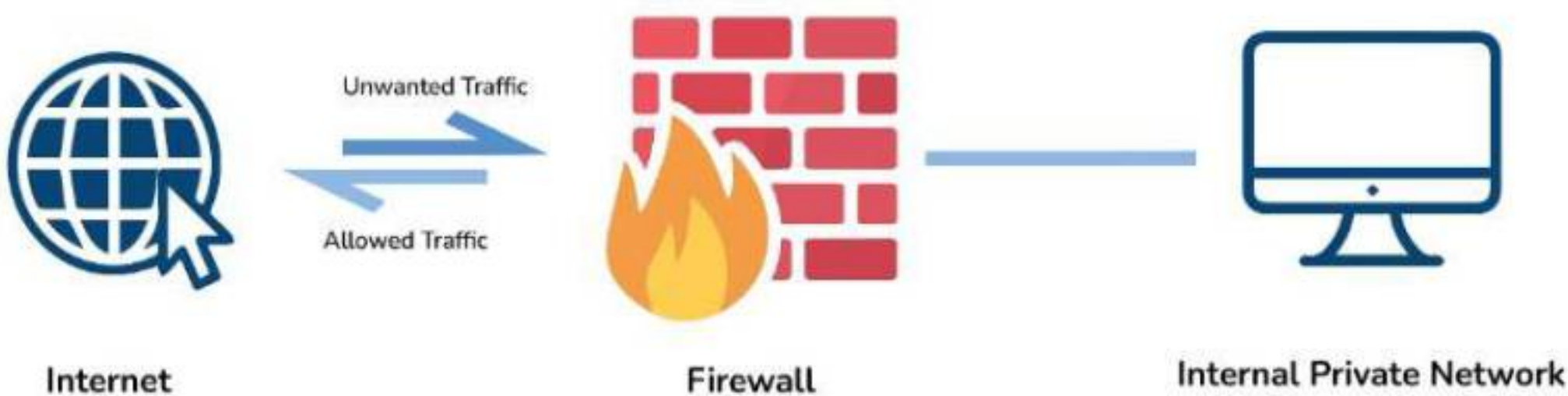
32. What is the difference between the ipconfig and the ifconfig?

ipconfig	ifconfig
Internet Protocol Configuration	Interface Configuration
Command used in Microsoft operating systems to view and configure network interfaces	Command used in MAC, Linux, UNIX operating systems to view and configure network interfaces
Used to get the TCP/IP summary and allows to changes the DHCP and DNS settings	

33. What is the firewall?

The firewall is a network security system that is used to monitor the incoming and outgoing traffic and blocks the same based on the firewall security policies. It acts as a wall between the internet (public network) and the networking devices (a private network). It is either a hardware device, software program, or a combination of both. It adds a layer of security to the network.

Firewall



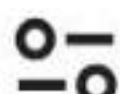
34. What are Unicasting, Anycasting, Multicasting and Broadcasting?

- **Unicasting:** If the message is sent to a single node from the source then it is known as unicasting. This is commonly used in networks to establish a new connection.
- **Anycasting:** If the message is sent to any of the nodes from the source then it is known as anycasting. It is mainly used to get the content from any of the servers in the Content Delivery System.
- **Multicasting:** If the message is sent to a subset of nodes from the source then it is known as multicasting. Used to send the same data to multiple receivers.
- **Broadcasting:** If the message is sent to all the nodes in a network from a source then it is known as broadcasting. DHCP and ARP in the local network use broadcasting.

35. What happens when you enter google.com in the web browser?

Below are the steps that are being followed:

- Check the browser cache first if the content is fresh and present in cache display the same.
- If not, the browser checks if the IP of the URL is present in the cache (browser and OS) if not then request the OS to do a DNS lookup using UDP to get the corresponding IP address of the URL from the DNS server to establish a new TCP connection.
- A new TCP connection is set between the browser and the server using three-way handshaking.
- An HTTP request is sent to the server using the TCP connection.
- The web servers running on the Servers handle the incoming HTTP request and send the HTTP response.
- The browser process the HTTP response sent by the server and may close the TCP connection or reuse the same for future requests.
- If the response data is cacheable then browsers cache the same.
- Browser decodes the response and renders the content.



- [Advanced Networking Interview Questions](#)

Basic Networking Interview Questions

Now, let's get started with the [list](#) of top Networking Interview Questions:

1. Name two technologies by which you would connect two offices in remote locations.

Two technologies that would connect two offices in remote locations are [VPN](#) and [Cloud computing](#).

2. What is internetworking?

Internetworking is a combination of two words, inter and networking which implies an association between totally different nodes or segments. This connection area unit is established through intercessor devices akin to routers or [gateways](#). The first term for associate degree internetwork was interconnected. This interconnection is often among or between public, private, commercial, industrial, or governmental networks.

Thus, associate degree internetwork could be an assortment of individual networks, connected by intermediate networking devices, that function as one giant network. Internetworking refers to the trade, products, and procedures that meet the challenge of making and administering Internet works.

3. Name of the software layers or User support layer in the OSI model.

- [Application layer](#)
- [Presentation layer](#)
- [Session layer](#)

4. Name the hardware layers or network support layers in the OSI model.

- [Network layer](#)
- [Datalink layer](#)
- [Physical layer](#)

5. Define HTTPS protocol?

The full form of [HTTPS](#) is a Hypertext transfer protocol secure. It is an advanced version of the [HTTP](#) protocol. Its port number is 443 by default. It uses [SSL/TLS](#) protocol for providing security.

6. Name some services provided by the application layer in the Internet model?

Some services provided by the application layer in the Internet model are as follows:

- Mail services
- Directory services
- File transfer
- Access management
- Network virtual terminal

7. In which OSI layer is the header and trailer added?

At the Data link layer trailer is added and at the [OSI model](#) layer 6,5,4,3 added header.

8. What happens in the OSI model, as a data packet moves from the lower to upper layers?

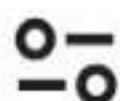
In the OSI model, as a data packet moves from the lower to upper layers, headers get removed.

9. What happens in the OSI model, as a data packet moves from the upper to lower layers?

In the OSI model, as a data packet moves from the upper to lower layers, headers are added. This header contains useful information.

10. What is a zone-based firewall?

A Zone-based firewall is an advanced method of [stateful firewall](#). In a stateful firewall, a stateful database is maintained in which the source [destination IP address](#), source port number, and destination port number are recorded. Due to this, only the traffic is allowed to pass if the traffic



header contains useful information.

10. What is a zone-based firewall?

A Zone-based firewall is an advanced method of stateful firewall. In a stateful firewall, a stateful database is maintained in which the source IP address, destination IP address, source port number, and destination port number are recorded. Due to this, only the replies are allowed i.e. if the traffic is Generated from inside the network then only the replies (of inside network traffic) coming from outside the network are allowed.

Cisco IOS [router](#) can be made firewall through two methods:

1. By using CBAC: create an access list and apply it to the interfaces keeping in mind what traffic should be allowed or denied and in what direction. This has an extra overhead for the administrator.
2. Using a Zone-based firewall.

For more details please refer [Zone-based firewall](#) article.

11. What is a server farm?

A server farm is a [set](#) of many servers interconnected together and housed within the same physical facility. A server farm provides the combined computing power of many servers by simultaneously executing one or more applications or services. A server farm is generally a part of an enterprise data center or a component of a supercomputer. A server farm is also known as a server cluster or computer ranch.

12. Name the three means of user authentication.

There is biometrics (e.g. a thumbprint, iris scan), a token, or a password. There is also two-level authentication, which employs two of those methods.

13. What is Confidentiality, Integrity & Availability?

The [CIA triad](#) can be broadly defined as:

Confidentiality – means information is not disclosed to unauthorized individuals, entities, or processes. For example, if we say I have a password for my Gmail account but someone saw it while I was doing login into my Gmail account. In that case, my password has been compromised and Confidentiality has been breached.

Integrity – means maintaining the accuracy and completeness of data. This means data cannot be edited in an unauthorized way. For example, if an employee leaves an organization then in that case data for that employee in all departments like accounts, should be updated to reflect the status to JOB LEFT so that data is complete and accurate in addition, this is only authorized persons should be allowed to edit employee data.

Availability – means information must be available when needed. For example, if one needs to access information about a particular employee to check whether an employee has outstood the number of leaves, that case, it requires collaboration from different organizational teams like network operations, [development](#) operations, incident response, and policy/change management. Denial of service attack is one of the factors that can hamper the availability of information.

14. What is VPN?

[VPN](#) stands for the virtual private network. A virtual private network (VPN) is a technology that creates a safe and encrypted connection over a less secure network, such as the Internet. A Virtual Private Network is a way to extend a private network using a public network such as the Internet. The name only suggests that it is a Virtual "private network" i.e. user can be part of a local network sitting at a remote location. It makes use of tunneling protocols to establish a secure connection.

15. What is Symmetric and Asymmetric Encryption?

- **Symmetric Key Encryption:** [Encryption](#) is a process to change the form of any message in order to protect it from reading by anyone. In Symmetric-key encryption the message is encrypted by using a key and the same key is used to decrypt the message which makes it easy to use but less secure. It also requires a safe method to transfer the key from one party to another.
- **Asymmetric Key Encryption:** Asymmetric Key Encryption is based on public and private key encryption techniques. It uses two different keys to encrypt and decrypt the message. It is more secure than the symmetric key encryption technique but is much slower. For more details please

[refer difference between symmetric and asymmetric encryption articles.](#)

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16. At what layer IPsec works?

An [IPsec](#) works on layer 3 of the OSI model.

17. What is a Tunnel mode?

This is a mode of data exchange wherein two communicating computers do not use IPsec themselves. Instead, the gateway that is connecting their LANs to the transit network creates a virtual tunnel that uses the IPsec protocol to secure all communication that passes through it. Tunnel mode is most commonly used between gateways, or at an end-station to a gateway, the gateway acting as a proxy for the hosts behind it. Tunnel mode is most commonly used to encrypt traffic between secure IPsec gateways, such as between the Cisco router and PIX Firewall

18. Define Digital Signatures?

As the name sounds are the new alternative to signing a document digitally. It ensures that the message is sent to the intended use without any tampering by any third party (attacker). In simple words, digital signatures are used to verify the authenticity of the message sent electronically.

OR

A digital signature is a mathematical technique used to validate the authenticity and integrity of a message, software, or digital document.

19. What is Authorization?

Authorization provides capabilities to enforce policies on network resources after the user has gained access to the network resources through authentication. After the authentication is successful, authorization can be used to determine what resources is the user allowed to access and the operations that can be performed.

20. What is the difference between IPS and a firewall?

The **Intrusion Prevention System** is also known as Intrusion Detection and Prevention System. It is a [network security](#) application that monitors network or system activities for malicious activity. The major functions of intrusion prevention systems are to identify malicious activity, collect information about this activity, report it, and attempt to block or stop it. Intrusion prevention systems are contemplated as augmentation of [Intrusion Detection Systems \(IDS\)](#), because both IPS and IDS operate network traffic and system activities for malicious activity. IPS typically records information related to observed events, notifies security administrators of important observed events, and produces reports. Many IPS can also respond to a detected threat by attempting to prevent it from succeeding. They use various response techniques, which involve the IPS stopping the attack itself, changing the security environment, or changing the attack's content.

A [firewall](#) is a network security device, either hardware or software-based, which monitors all incoming and outgoing traffic, and based on a defined set of security rules it accepts, rejects, or drops that specific traffic.



Networking Interview Questions

Intermediate Networking Interview Questions





Networking Interview Questions

Intermediate Networking Interview Questions

21. What is IP Spoofing?

IP Spoofing is essentially a technique used by [hackers](#) to gain unauthorized access to Computers. Concepts of IP Spoofing were initially discussed in academic circles as early as 1980. IP Spoofing types of attacks had been known to Security experts on the theoretical level. It was primarily theoretical until Robert Morris discovered a security weakness in the [TCP protocol](#) known as sequence prediction. Occasionally IP spoofing is done to mask the origins of a Dos attack. In fact, [Dos attacks](#) often mask the actual IP addresses from where the attack has originated from.

22. What is the meaning of threat, vulnerability, and risk?

Threats are anything that can exploit a vulnerability accidentally or intentionally and destroy or damage an **asset**. An asset can be anything people, property, or information. The asset is what we are trying to protect and a threat is what we are trying to protect against. **Vulnerability** means a gap or weakness in our protection efforts.

Risk is nothing but an intersection of assets, threats, and vulnerability.

$$A+T+V = R$$

23. What is the main purpose of a DNS server?

DNS stands for [Domain Name Server](#). It translates Internet domains and hostnames to IP addresses and vice versa. DNS technology allows typing names into your Web browsers and your computer to automatically find that address on the Internet. A key element of the DNS is a worldwide collection of DNS servers. It has the responsibility of assigning domain names and mapping those names to Internet resources by designating an authoritative_name server for each domain. The Internet maintains two main namespaces like Domain Name hierarchy and Internet protocol address space.

24. What is the protocol and port no of DNS?

Protocol - [TCP/UDP](#)

Port number- 53

25. What is the position of the transmission media in the OSI model?

In the OSI model, transmission media supports layer-1(Physical layer).

26. What is the importance of twisting in the twisted-pair cable?

The [twisted-pair cable](#) consists of two insulated copper wires twisted together. The twisting is important for minimizing electromagnetic radiation and external interference.

27. What kind of error is undetectable by the checksum?

In [checksum](#), multiple-bit errors can not be undetectable.

28. Which multiplexing technique is used in the Fiber-optic links?

The wavelength division multiplexing is commonly used in fiber optic links.

29. What are the Advantages of Fiber Optics?

The advantages of Fiber Optics are mentioned below:

- Bandwidth is above copper cables.
- Less [power](#) loss and allows data transmission for extended distances.
- The optical cable is resistant to electromagnetic interference.
- Fiber cable is sized 4.5 times which is best than copper wires.
- As the cable is lighter, and thinner, in order that they use less area as compared to copper wires.
- Installation is extremely easy thanks to less weight.
- Optical fiber cable is extremely hard to tap because they don't produce electromagnetic energy. These optical fiber cables are very secure for transmitting data.
- This cable opposes most acidic elements that hit copper wires also are flexible in nature.
- Optical fiber cables are often made cheaper than equivalent lengths of copper wire.
- Light has the fastest speed within the universe, much a lot faster signals.

• Fiber optic cables allow much more cable than copper twisted-pair cables.

• Fiber optic cables have how more bandwidth than copper twisted-pair cables.

The wavelength division multiplexing is commonly used in fiber optic links.

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- Light has the fastest speed within the universe, such a lot faster signals.
- Fiber optic cables allow much more cable than copper twisted-pair cables.
- Fiber optic cables have how more bandwidth than copper twisted-pair cables.

30. Which of the multiplexing techniques are used to combine analog signals?

To combine analog signals, commonly FDM(Frequency division multiplexing) and WDM (Wavelength-division multiplexing) are used.

31. Which of the multiplexing techniques is used to combine digital signals?

To combine digital signals, time division multiplexing techniques are used.

32. Can IP Multicast be load-balanced?

No, The IP multicast multipath command load splits the traffic and does not load balance the traffic. Traffic from a source will use only one path, even if the traffic far outweighs traffic from other sources.

33. What is CGMP(Cisco Group Management Protocol)?

CGMP is a simple protocol, the routers are the only devices that are producing CGMP messages. The switches only listen to these messages and act upon them. CGMP uses a well-known destination **MAC address (0100.0cdd.dddd)** for all its messages. When switches receive frames with this destination address, they flood it on all their interfaces Bluetoothso all switches in the network will receive CGMP messages.

Within a CGMP message, the two most important items are:

- Group Destination Address (GDA)
- Unicast Source Address (USA)

The group destination address is the multicast group MAC address, and a unicast source address is the MAC address of the host (receiver).

34. What is Multicast?

Multicast is a method of group communication where the sender sends data to multiple receivers or nodes present in the network simultaneously. Multicasting is a type of one-to-many and many-to-many communication as it allows sender or senders to send data packets to multiple receivers at once across LANs or WANs. This process helps in minimizing the data frame of the network. For more details please read Multicasting in computer network article.

35. What is the difference between Bluetooth and wifi?

Bluetooth	Wifi
<u>Bluetooth</u> has no full form.	While <u>Wifi</u> stands for Wireless Fidelity.
It requires a Bluetooth adapter on all devices for connectivity.	Whereas it requires a wireless adapter Bluetooth for all devices and a wireless router for connectivity.
Bluetooth consumes low power.	while it consumes high power.
The security of BlueTooth is less in comparison to the number of wifi.	While it provides better security than BlueTooth.
Bluetooth is less flexible means these limited users are supported.	Whereas wifi supports a large number of users.

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The radio signal range of BlueTooth is ten meters.	Whereas in wifi this range is a hundred meters.
Bluetooth requires low bandwidth.	While it requires high bandwidth.

36. What is a reverse proxy?

Reverse Proxy Server: The job of a reverse proxy server is to listen to the request made by the client and redirect to the particular web server which is present on different servers. This is also used to restrict the access of the clients to the confidential data residing on particular servers. For more details please refer to [what is proxy server](#) article.

37. What is the role of address in a packet traveling through a datagram network?

The address field in a datagram network is end-to-end addressing.

38. Can a routing table in the datagram network have two entries with the same destination address?

No. [routing](#) tables in the datagram network have two entries with the same destination address, not possible because the destination address or receiver address is unique in the datagram network.

39. What kind of arithmetic is used to add data items in checksum calculation?

To add data items in checksum calculations, one's complement arithmetic is used.

40. Define piggybacking?

[Piggybacking](#) is used to improve the efficiency of the bidirectional protocols. When a frame is carrying data from A to B, it can also carry control information about arrived (or lost) frames from B; when a frame is carrying data from B to A, it can also carry control information about the arrived (or lost) frames from A.

Advanced Networking Interview Questions

41. What are the advantages and disadvantages of piggybacking?

Advantages of Piggybacking:

The major advantage of piggybacking is the better use of available channel bandwidth.

Disadvantages of Piggybacking:

The major disadvantage of piggybacking is additional complexity and if the data link layer waits too long before transmitting the acknowledgment, then re-transmission of the frame would take place.

42. Which technique is used in byte-oriented protocols?

Byte stuffing is used in byte-oriented protocols. A special byte is added to the data section of the frame when there is a character with the same pattern as the flag.

43. Define the term OFDM?

[Orthogonal Frequency Division Multiplexing \(OFDM\):](#)

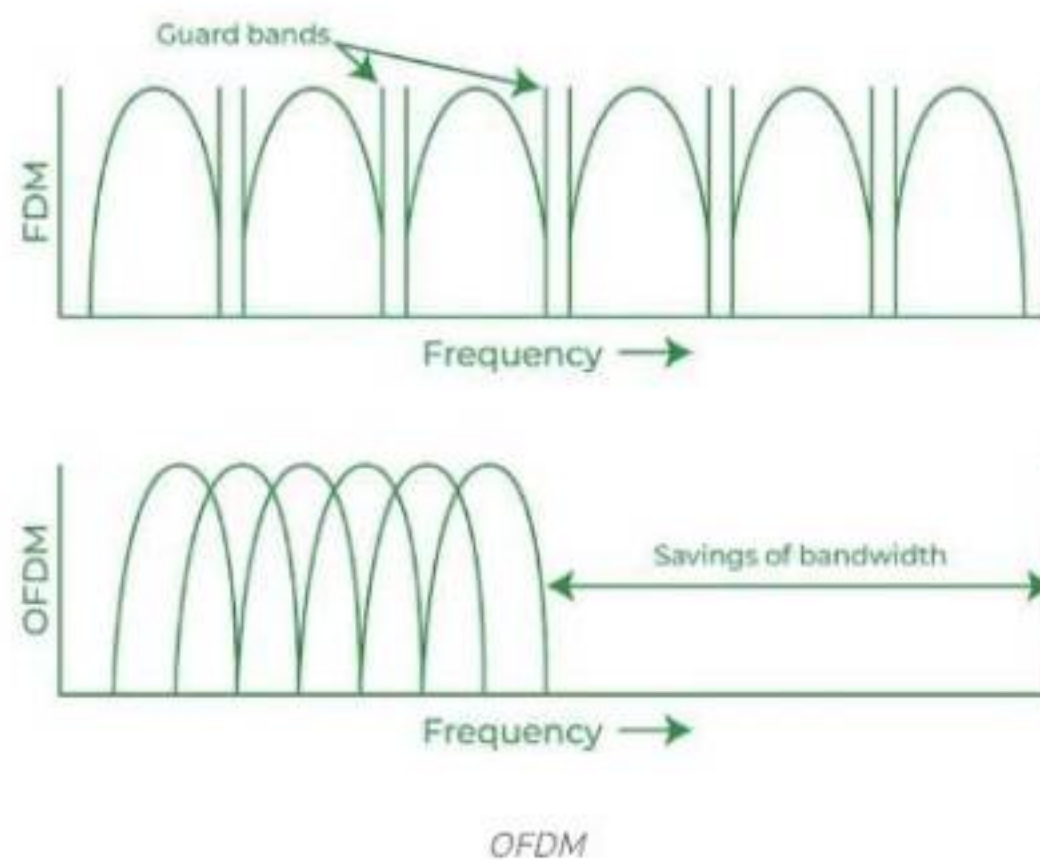
It is also the multiplexing technique that is used in an analog system. In OFDM, the Guard band is not required and the spectral efficiency of OFDM is high which oppose to the FDM. In OFDM, a Single data source attaches all the sub-channels

name when there is a character with the same pattern as the tag.

43. Define the term OFDM?

Orthogonal Frequency Division Multiplexing (OFDM):

It is also the multiplexing technique that is used in an analog system. In OFDM, the Guard band is not required and the spectral efficiency of OFDM is high which oppose to the FDM. In OFDM, a Single data source attaches all the sub-channels.



44. What is a transparent bridge?

Transparent Bridge:

A transparent bridge automatically maintains a routing table and updates tables in response to maintaining changing topology. The transparent bridge mechanism consists of three mechanisms:

- Frame forwarding
- Address Learning
- Loop Resolution

The Transparent bridge is easy to use. Install the bridge and no software changes are needed in the hosts. In all the cases, transparent bridges flooded the broadcast and multicast frames.

45. What is the minimum size of the icmpV4 packet what is the maximum size of the icmpv4 packet?

- Minimum size ICMPv4 packet = 28 bytes
- Maximum size ICMPv4 packet = 2068 bytes

46. Why do we OSPF a protocol that is faster than our RIP?

OSPF stands for Open Shortest Path First which uses a link-state routing algorithm. This protocol is faster than RIP because:

- Using the link-state information which is available in routers, it constructs the topology of Bluetooth which Bluetooth the topology determines the routing table for routing decisions.
- It supports both variable-length subnet masking and classless inter-domain routing addressing models.
- Since it uses Dijkstra's algorithm, it computes the shortest path tree for each route.
- OSPF (Open Shortest Path First) is handling the error detection by itself and it uses multicast addressing for routing in a broadcast domain

47. What are the two main categories of DNS messages?

The two categories of DNS messages are queries and replies.

48. Why do we need the pop3 protocol for e-mail?

Need of POP3: The Post Office Protocol (POP3) is the most widely used protocol and is supported by most email clients. It provides a convenient and standard way for users to access mailboxes and download messages. An important advantage of this is that the mail messages get delivered to the client's PC and they can be read with or without accessing the web.

49. Define the term Jitter?

Jitter is a "packet delay variance". It can simply mean that jitter is considered a problem when different packets of data face different delays in a network and the data at the receiver application is time-sensitive, i.e. audio or video data. Jitter is measured in milliseconds(ms). It is defined as an interference in the normal order of sending data packets.

50. Why Bandwidth is important to network performance parameters?

Bandwidth is characterized as the measure of data or information that can be transmitted in a fixed measure of time. The term can be used in **Open In App** contexts with two distinctive estimating

interference in the normal order of sending data packets.

50. Why Bandwidth is important to network performance parameters?

Bandwidth is characterized as the measure of data or information that can be transmitted in a fixed measure of time. The term can be used in two different contexts with two distinctive estimating values. In the case of digital devices, the bandwidth is measured in bits per second(bps) or bytes per second. In the case of analog devices, the bandwidth is measured in cycles per second, or Hertz (Hz). Bandwidth is only one component of what an individual sees as the speed of a network. True internet speed is actually the amount of data you receive every second and that has a lot to do with latency too.

51. How do I Identify When an IP Address is Private or Public?

You can identify private IP addresses by checking if they fall within the reserved ranges (e.g., 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16)

52. How To Get an IP Address from Domain Name?

Answer: We can get an IP address from a domain name using ping commands and nslookup command.For this, use command-line tools like PING or nslookup to get the IP address. Run the commands "PING example.com" or "nslookup example.com" on command prompt or terminal window.

53. Which Diffie Hellman Group is Most Secure?

The most secure Diffie-Hellman group is currently considered to be Group 24 (2048-bit ECP) or higher, offering stronger encryption and resistance to attacks. Apart from that the security of a Diffie-Hellman (DH) group depends on the size and type of the underlying prime numbers or elliptic curves used.

54. How Flow Control is Achieved in TCP?

In computer networks, reliable data delivery is important. The Transmission Control Protocol guarantees in-order and error-free data transfer using flow control. This is to prevent the sender from flooding the receiver so as to make sure it can work efficiently in turn.TCP utilizes a sliding window protocol for flow control. The receiver advertises a window size, indicating the number of bytes its buffer can hold. The sender transmits data segments up to this advertised window

55. How To Find Your Port Number ?

We can find port number using command line Tool, and using resource monitor. By utilizing the tools like 'Netstat' we can troubleshoot and monitor our system and network, and also gain the insights into network security, and identify any processes using specific ports. It will help us in managing and securing our system efficiently.

Conclusion

Getting ready for a networking interview can be tough, but having the right questions and answers at your fingertips makes it easier. We've put together over 50 common networking interview questions and answers to help you prepare. Whether you're just starting or have years of experience, these questions cover the important topics you'll need to know. Take the time to go through them, understand the basics, and think about how you'd handle real-world situations.

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Here is a cheat sheet of computer networks that is given below:

Network Topologies:

Name	Description
Bus Topology	A bus topology, also called a line topology, is a type of network topology in which all network devices are connected through a central RJ-45 network cable or coaxial cable.
Ring Topology	A ring topology is a type of network topology in which each device is connected to two other devices on either side using RJ-45 or coaxial cables.
Star Topology	A star topology is a network topology in which each element of the network is physically connected to a central node such as a router, hub, or switch. In a star topology, hubs act as servers, and connecting nodes act as clients.
Mesh Topology	In a mesh topology, each node is connected to at least one other node and often to multiple nodes. Each node can send and receive messages from other nodes.
Tree Topology	A tree topology is a hybrid network topology in which star networks are interconnected by bus networks. Tree networks are hierarchical and each node can have any number of child nodes.
Hybrid Topology	A hybrid topology is a type of network topology that uses two or more different network topologies. These topologies can include mixed bus topologies, mesh topologies, ring topologies, star topologies, and tree topologies.

Types of Network:

Network Type	Description
PAN	Personal Network is a network consisting of only a small number of devices owned by an individual.
LAN	A local area network is a network that covers a small area (for example, a company's network).
WAN	A wide Area Network is a network that includes many devices and covers a large area. Usually collectively owned.
MAN	MAN stands for Metropolitan Area Network. It is a computer network that connects a findnumber of LANs to form a larger network so that the computer resources can be shared.

TCP/IP Model and OSI Model:

TCP/IP	OSI Model	Protocols
Application Layer	Application Layer	DNS, DHCP, FTP, HTTPS, IMAP, LDAP, NTP,POP3, RTP, RTSP, SSH, SIP, SMTP, SNMP, Telnet, TFTP
	Presentation Layer	JPEG, MIDI, MPEG, PICT, TIFF
	Session Layer	NetBIOS, NFS, PAP, SCP, SQL, ZIP
Transport Layer	Transport Layer	TCP, UDP
Internet Layer	Network Layer	ICMP, IGMP, IPsec, IPv4, IPv6, IPX, RIP
Link Layer	Data Link Layer	ARP, ATM, CDP, FDDI, Frame Relay, HDLC, MPLS, PPP, STP, Token Ring
Physical Layer		Ethernet, DSL, ISDN, 802.11, Wi-Fi

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<u>Application Layer</u>	<u>Application Layer</u>	DNS, DHCP, FTP, HTTPS, IMAP, LDAP, NTP,POP3, RTP, RTSP, SSH, SIP, SMTP, SNMP, Telnet, TFTP
	<u>Presentation Layer</u>	JPEG, MIDI, MPEG, PICT, TIFF
	<u>Session Layer</u>	NetBIOS, NFS, PAP, SCP, SQL, ZIP
<u>Transport Layer</u>	<u>Transport Layer</u>	TCP, UDP
<u>Internet Layer</u>	<u>Network Layer</u>	ICMP, IGMP, IPsec, IPv4, IPv6, IPX, RIP
<u>Link Layer</u>	<u>Data Link Layer</u>	ARP, ATM, CDP, FDDI, Frame Relay, HDLC, MPLS, PPP, STP, Token Ring
	<u>Physical Layer</u>	Bluetooth, Ethernet, DSL, ISDN, 802.11 Wi-Fi

Computer Network Protocols:

Network Protocol	Description	Port number of protocol
<u>Ethernet</u>	A family of protocols that specify how devices on the same network segment format and transmit data.	44818, 2222
<u>Wi-Fi or WLAN</u>	A family of protocols that deal with wireless transmission.	-
<u>TCP</u>	Splits data into packets (reassembles later). Error checking is also included, as the acknowledgment is expected to be sent within a specified timeframe.	22
<u>UDP</u>	User Datagram Protocol	4096-65535
<u>IP</u>	Every device has an IP address. Packets are "addressed" to ensure they reach the correct user.	-
<u>HTTP</u>	Used to access web pages from a web server.	80
HTTP'S	uses encryption to protect data.	443
<u>FTP</u>	File Transfer Protocol: Handles file uploads and downloads, transferring data and programs.	21
<u>SMTP</u>	SMTP server has a database of user email addresses. Internet Message Access Protocol: Handles incoming mail.	587
<u>IMAP</u>	Internet Message Access Protocol: Process incoming mail.	993
<u>ARP</u>	ARP finds a host's hardware address (also known as MAC (Media Access Control) address) based on its known IP address.	-
<u>DNS</u>	DNS is the host name for the IP address translation service. DNS is a distributed database implemented on a hierarchy of name servers. It is an application layer protocol for messaging between clients and servers.	53
<u>FTPS</u>	FTPS is known as FTP SSL which refers to <u>File Transfer Protocol (FTP)</u> , over <u>Secure Sockets Layer (SSL)</u> , which is more secure from FTP. FTPS also called as File Transfer Protocol Secure.	21

BOOTP	Bootstrap Protocol (BOOTP) is a network protocol used by network management to assign IP addresses to each member of that network in order to join other network devices through a main server.	67
DHCP	Dynamic Host Configuration Protocol (DHCP) is an application layer protocol. DHCP is based on a client-server model, based on discoveries, offers, requests, and ACKs.	68
NAT	Network Address Translation (NAT) is the process of translating one or more local IP addresses into one or more global IP addresses, or vice versa, in order to provide Internet access to local hosts.	5351
PPP	Point-to-Point Protocol (PPP) is basically an asymmetric protocol suite for various connections or links without framing. H. Raw bit pipe. PPP also expects other protocols to establish connections, authenticate users, and carry network layer data as well.	1994
RIP	Routing Information Protocol (RIP) is a dynamic routing protocol that uses hop count as a routing metric to find the best path between source and destination networks.	520
OSPF	Open Shortest Path First (OSPF) is a link-state routing protocol used to find the best path between a source and destination router using its own shortest path first).	89
EIGRP	Enhanced Interior Gateway Routing Protocol (EIGRP) is a dynamic routing protocol used to find the best path and deliver packets between any two Layer 3 devices.	88
BGP	Border Gateway Protocol (BGP) is a protocol used to exchange Internet routing information and is used between ISPs in different ASes.	179
STP	Spanning Tree Protocol (STP) is used to create a loop-free network by monitoring the network, tracking all connections, and shutting down the least redundant connections.	0 to 255
RARP	RARP, stand for Reverse Address Resolution Protocol, is a computer network-based protocol used by client computers to request IP addresses from a gateway server's Address Resolution Protocol table or cache.	-
LAPD	The D-channel LAPD or Link Access Protocol is basically the Layer 2 protocol normally required for the ISDN D-channel. It is derived from the LAPB (Link Access Protocol Balanced) protocol.	-
IPsec	IP Security (IPSec) is a standard suite of Internet Engineering Task Force (IETF) protocols between two communication points on IP networks to provide data authentication, integrity, and confidentiality. It also defines encrypted, decrypted, and authenticated packets.	4500
ASCII	ASCII (American Standard Code for Information Interchange) is the standard character encoding used in telecommunications. The ASCII representation "ask-ee" is strictly a 7-bit code based on the English alphabet. ASCII codes are used to represent alphanumeric data.	9500
EBCDIC	EBCDIC (Extended Binary Encoded Decimal Interchange Code) (pronounced "ehb-suh-dik" or "ehb-kuh-dik") is an alphanumeric binary code developed by IBM to run large-scale computer systems .	-
X.25 PAD	X.25 is an International Telecommunication Union Telecommunication Standardization Sector (ITU-T) protocol standard simply for Wide Area Network (WAN) , communications that basically describes how the connections among user devices and network devices are established and maintained.	-
HDLC	High-Level Data Link Control (HDLC) commonly uses the term "frame" to denote units or logs of units of data that are frequently transmitted or transmitted from one station to another, express. Each frame on the link must start and end with a flag.	-

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SLIP	SLIP stands for Serial Line Internet Protocol. It is a TCP/IP implementation which was described under RFC 1055 (Request for Comments).	-
LAP	Link Access Procedure (LAP) is basically considered as an ITU family of Data Link Layer (DLL) protocols that are subsets of High-Level Data Link Control (HDLC) . LAP is particularly derived from IBM's System Development Life Cycle (SDLC) .	-
NCP	Network Control Protocol (NCP) is a set of protocols that are part of Point-to-Point Protocol (PPP).	524
Mobile IP	Mobile IP is a communication protocol (created by extending the Internet Protocol, IP) that allows a user to move from one network to another using the same her IP address.	434
VOIP	Voice over Internet Protocol (VoIP), is a technology that allowing you to make voice calls over a broadband Internet connection instead of an analog (regular) phone line. Some VoIP services allow you to call people using the same service, but others may allow you to call anyone.	5060
LDAP	Lightweight Directory Access Protocol (LDAP) is an internet protocol works on TCP/IP, used to access information from directories. LDAP protocol is basically used to access an active directory.	389
GRE	GRE or Generic Routing Encapsulation is a tunneling protocol developed by Cisco. It encapsulates IP packets i.e. deliverable inner packets into outer packets.	47
AH	The HTTP headers Authorization header is a request type header that used to contains the credentials information to authenticate a user through a server. If the server responds with 401 Unauthorized and the WWW-Authenticate header not usually.	51
ESP	Encapsulation security payload, also abbreviated as ESP plays a very important role in network security. ESP or Encapsulation security payload is an individual protocol in IPsec.	500
NNTP	Network News Transfer Protocol (NNTP) is the underlying protocol of UseNet, which is a worldwide discussion system which contains posts or articles which are known as news.	119
RPC-DCOM	DCOM- Distributed Component Object Model – helps remote object via running on a protocol known as the Object Remote Procedure Call (ORPC).	-
IRC	Internet Relay Chat (IRC) is an Internet application that was developed by Jakko Oikarinen in Finland. Chat is the most convenient immediate way to communicate with others via Internet.	6667

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IRC

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IEEE Standards:

Standards	Description
IEEE 802	LAN/MAN
IEEE 802.1	LAN/MAN Bridging and management
IEEE 802.1s	Multiple spanning tree
IEEE 802.1w	Rapid reconfiguration of spanning tree
IEEE 802.1x	Port-based network access control
IEEE 802.2	Logical Link Control (LLC).
IEEE 802.3	CSMA/CD access method (Ethernet).
IEEE 802.3ae	10 Gigabit Ethernet
IEEE 802.4	Token passing bus access method and Physical layer specifications
IEEE 802.5	Token Ring access method and Physical layer specifications
IEEE 802.6	Distributed Queue Dual Bus (DQDB) access method and Physical layer specifications (MAN).
IEEE 802.7	Broadband LAN
IEEE 802.8	Fiber Optic
IEEE 802.9	Isochronous LANs (standard withdrawn)
IEEE 802.10	Interoperable LAN/MAN Security
IEEE 802.11	Wireless LAN MAC and Physical layer specifications
IEEE 802.11a	Wireless with speed upto 54 Mbps
IEEE 802.11b	Wireless with speed upto 11 Mbps
IEEE 802.11g	Wireless with speed upto 54 Mbps
IEEE 802.11n	Wireless with speed upto 600 Mbps
IEEE 802.12	Demand-priority access method, physical layer and repeater specifications
IEEE 802.13	Not used
IEEE 802.14	Cable modems (proposed standard was withdrawn).
IEEE 802.15	Wireless Personal Area Network (WPAN).
IEEE 802.16	Wireless Metropolitan Area Network (Wireless MAN).
IEEE 802.17	Resilient Packet Ring (RPR) Access

Networking Devices:

Device	Description
Client	Any device, such as a workstation, laptop, tablet, or smartphone, that is used to access a network.

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Cables in Networking Devices:

Ethernet Standards (IEEE)	Data Rate	Cable Fiber Type	Maximum Distance (IEEE)
Ethernet (10Base-FL)	10 Mbps	50m or 62.5um Multimode @ 850nm	2km
Fast Ethernet (100Base-FX)	100 Mbps	50m or 62.5um Multimode @ 1300nm	2km
Fast Ethernet (100Base-SX)	100 Mbps	50m or 62.5um Multimode @ 850nm	300m
Gigabit Ethernet (1000Base-SX)	1000 Mbps	50m Multimode @ 850nm	550m
Gigabit Ethernet (1000Base-SX)	1000 Mbps	62.5um Multimode @ 850nm	220m
Gigabit Ethernet (1000Base-LX)	1000 Mbps	50m or 62.5um Multimode @ 1300nm	550m
Gigabit Ethernet (1000Base-LX)	1000 Mbps	9um Singlemode @1310nm	5km
Gigabit Ethernet (1000Base-LH)	1000 Mbps	9um Singlemode @1550nm	70km

Types of Ethernet Networks:

Speed	Common Name	Informal IEEE Standard Name	Formal IEEE Standard Name	Cable Type, Maximum Length
10 Mbps	Ethernet	10BASE-T	802.3	Copper, 100 m
100 Mbps	Fast Ethernet	100BASE-T	802.3u	Copper, 100 m
1000 Mbps	Gigabit Ethernet	1000BASE-LX	802.3z	Fiber, 5000 m
1000 Mbps	Gigabit Ethernet	1000BASE-T	802.3ab	Copper, 100 m
10 Gbps	10 Gig Ethernet	10GBASE-T	802.3an	Copper, 100 m

Types of Network Connections:

Type	Description
Internet	A network of millions of interconnected and cooperatively connected computers is called the Internet. Internet includes people, resources and means of collaboration
Intranet	It is an internal private network built within an organization using Internet and World Wide Web standards and products that provides access to corporate information for the organization's employees.
Extranet	This is a type of network that allows external users to access an organization's intranet.

Transmission Media:

- Guided Media:

Type of media	Description
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Extranet

This is a type of network that allows external users to access an organization's intranet.

Transmission Media:

Guided Media:

Type of media	Description
Twisted Pair Cable	It is a superimposed winding of two separately insulated conductors. As a rule, several such pairs are grouped together in a protective cover. They are the most widely used transmission media.
Coaxial Cable	It has a PVC or Teflon insulating layer and an outer plastic sheath containing two parallel conductors, each with a separate conformal protective cover.
Optical Fiber Cable	It uses the concept of light reflection through a glass or plastic core. The core is surrounded by a less dense glass or plastic shell called the cladding. Used to transfer large amounts of data.
Stripline	Stripline is a transverse electromagnetic (TEM) transmission line medium invented by Robert M. Barrett at the Air Force Cambridge Research Center in the 1950s. Stripline is the earliest form of planar transmission line.
Microstripline	Conductive material is separated from the ground plane by a dielectric layer.

Unguided Media:

Type of media	Description
Radio waves	These are easy to generate and can penetrate buildings. There is no need to align the transmit and receive antennas. Frequency Range: 3kHz - 1GHz AM radios, FM radios, and cordless phones use radio waves for transmission.
Microwaves	Multiplexer types: line-of-sight transmission. H. Transmitting and receiving antennas should be placed properly. The distance a signal travels is directly proportional to the height of the antenna. Frequency Range: 1GHz - 300GHz They are mainly used for mobile telephony and television distribution.
Infrared	Infrared is used for short distance communication. Obstacles cannot be penetrated. This prevents interference between systems. Frequency Range: 300GHz - 400THz It is used in TV remote controls, wireless mice, keyboards, printers, etc.

Types of Multiplexers:

Type	Description
Frequency Division Multiplexing (FDM)	The frequency spectrum is divided into logical channels and each user has exclusive access to his channel. It transmits signals in several different frequency ranges and multiple video channels over a single cable. Each signal is modulated onto a different carrier frequency and the carrier frequencies are separated by guard bands.
Time Division Multiplexing (TDM)	Each user gets full bandwidth for a short period of time on a regular basis. The entire channel is dedicated to her one user, but only for a short time.
Wavelength Division Multiplexing	This is the same as FDM but applied to fiber, with the difference that here the operating frequency is much higher, actually in the optical range. Due to its extremely high bandwidth, fiber optic has great potential.

Collision Detection:

Type	Description
Carrier Sense Multiple Access with Collision Detection (CSMA/CD)	In this method, after sending a frame, the station monitors the media to see if the transmission was successful. If successful, the transmission is terminated, otherwise the frame is retransmitted.

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The basic idea behind CSMA/CA is that station must be able to receive

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• Or the capacity is divided between signals traveling in both directions.

Classes in Computer Networking:

CLASS	LEADING BITS	NET ID BITS	HOST ID BITS	NO. OF NETWORKS	ADDRESSES PER NETWORK	START ADDRESS	1st Usable IP Address
CLASS A	0	8	24	2 ⁷ (128)	2 ²⁴ 2 (16,777,216)	0.0.0.0	127.255.255.254
CLASS B	10	16	16	2 ¹⁴ 2 (16,384)	2 ¹⁶ 2 (65,536)	128.0.0.0	191.255.255.254
CLASS C	110	24	8	2 ²¹ 2 (2,097,152)	2 ⁸ 2 (256)	192.0.0.0	223.255.255.254
CLASS D	1110	NOT DEFINED	NOT DEFINED	NOT DEFINED	NOT DEFINED	224.0.0.0	239.255.255.254
CLASS E	1111	NOT DEFINED	NOT DEFINED	NOT DEFINED	NOT DEFINED	240.0.0.0	255.255.255.254

Subnet Address or Subnet ID Using/16Prefix	1st Usable IP Address	Last usable IP Address	Broadcast Address
120.0.0.0/24	120.0.0.1	120.0.255.254	120.0.255.255
120.1.0.0/24	120.1.0.1	120.1.255.254	120.1.255.255
120.2.0.0/24	120.2.0.1	120.2.255.254	120.2.255.255
120.3.0.0/24	120.3.0.1	120.3.255.254	120.3.255.255
120.100.0.0/24	120.100.0.1	120.100.255.254	120.100.255.255
120.101.0.0/24	120.101.0.1	120.101.255.254	120.101.255.255
120.200.0.0/24	120.200.0.1	120.200.255.254	120.200.255.255
120.201.0.0/24	120.201.0.1	120.201.255.254	120.201.255.255
120.253.0.0/24	120.253.0.1	120.253.255.254	120.253.255.255
120.254.0.0/24	120.254.0.1	120.254.255.254	120.254.255.255
120.255.0.0/24	120.255.0.1	120.255.255.254	120.255.255.255

Subnetting:

Private IP Address with Subnet Mask	Private IP Range	Private IP Range denoted in CIDR
10.0.0.0 255.0.0.0	10.0.0.0 to 10.255.255.255	10.0.0.0/8
172.16.0.0 255.240.0.0	172.16.0.0 to 172.31.255.255	172.16.0.0/12
192.168.0.0 255.255.0.0	192.168.0.0 to 192.168.255.255	192.168.0.0/16

CIDR	SUBNET MASK	WILDCARD MASK	# OF IP ADDRESSES	# OF USABLE IP ADDRESSES
/32	255.255.255.255	0.0.0.0	1	1
/31	255.255.255.254	0.0.0.1	2	2*
/30	255.255.255.252	0.0.0.3	4	2

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CIDR	SUBNET MASK	WILDCARD MASK	# OF IP ADDRESSES	# OF USABLE IP ADDRESSES
/32	255.255.255.255	0.0.0.0	1	1
/31	255.255.255.254	0.0.0.1	2	2*
/30	255.255.255.252	0.0.0.3	4	2
/29	255.255.255.248	0.0.0.7	8	6
/28	255.255.255.240	0.0.0.15	16	14
/27	255.255.255.224	0.0.0.31	32	30
/26	255.255.255.192	0.0.0.63	64	62
/25	255.255.255.128	0.0.0.127	128	126
/24	255.255.255.0	0.0.0.255	256	254
/23	255.255.254.0	0.0.1.255	512	510
/22	255.255.252.0	0.0.3.255	1,024	1,022
/21	255.255.248.0	0.0.7.255	2,048	2,046
/20	255.255.240.0	0.0.15.255	4,096	4,094
/19	255.255.224.0	0.0.31.255	8,192	8,190
/18	255.255.192.0	0.0.63.255	16,384	16,382
/17	255.255.128.0	0.0.127.255	32,768	32,766
/16	255.255.0.0	0.0.255.255	65,536	65,534
/15	255.254.0.0	0.1.255.255	131,072	131,070
/14	255.252.0.0	0.3.255.255	262,144	262,142
/13	255.248.0.0	0.7.255.255	524,288	524,286
/12	255.240.0.0	0.15.255.255	1,048,576	1,048,574
/11	255.224.0.0	0.31.255.255	2,097,152	2,097,150
/10	255.192.0.0	0.63.255.255	4,194,304	4,194,302
/9	255.128.0.0	0.127.255.255	8,388,608	8,388,606
/8	255.0.0.0	0.255.255.255	16,777,216	16,777,214
/7	254.0.0.0	1.255.255.255	33,554,432	33,554,430
/6	252.0.0.0	3.255.255.255	67,108,864	67,108,862
/5	248.0.0.0	7.255.255.255	134,217,728	134,217,726
/4	240.0.0.0	15.255.255.255	268,435,456	268,435,454
/3	224.0.0.0	31.255.255.255	536,870,912	536,870,910
/2	192.0.0.0	63.255.255.255	1,073,741,824	1,073,741,822
/1	128.0.0.0	127.255.255.255	2,147,483,648	2,147,483,646
/0	0.0.0.0	255.255.255.255	4,294,967,296	4,294,967,294

Methods of Network Security:

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/3	224.0.0.0	31.255.255.255	536,870,912	536,870,910
/2	192.0.0.0	63.255.255.255	1,073,741,824	1,073,741,822
/1	128.0.0.0	127.255.255.255	2,147,483,648	2,147,483,646
/0	0.0.0.0	255.255.255.255	4,294,967,296	4,294,967,294

Methods of Network Security:

Method	Description
<u>Authentication</u>	Verify a user's identity, usually by asking them to enter a password or biometric identifier.
<u>Encryption</u>	Encrypt data with a key, that is, the same key is required to decrypt the data. This is how HTTPS works.
<u>Firewalls</u>	Protect the network from unauthorized access.
<u>MAC Address Filtering</u>	Allow devices to access or be prevented from accessing the network based on their physical address embedded in the device's network adapter.

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